

The Computational Mesoscale

Compute, Transport, and Memory Between Neural Execution and Model Behavior

Ivan GAN

ivan@ivan.icu www.ivan.icu github.com/ivanicu

Abstract

A neural architecture constrains more than which functions can be learned. It sets the rules by which information can be *computed*, *transported*, and *remembered*. Training then selects a particular organization of these three causal roles. Two models can therefore match behavior while differing in what they transform, where intermediate results become available, what distinctions persist, and where failure pressure enters. We call this missing causal level *the computational mesoscale* (MESO), and a scoped learned organization at that level a *MesoMachine*.

The compute–transport–memory (CTM) decomposition is a role system, not a partition into modules: one physical process may transform a variable, move it between computational sites, and establish a state that later computation can read. A model specification supplies a *MesoGenome*—its typed CTM syntax, including primitive operations, dependency paths, routing, persistence, addressing, cuts, sharing, position, and resources. A *MesoGene* is one declared editable factor inside that genome. Training conditions and learning dynamics induce a distribution over accessible implementation organizations:

$$(MesoGenome, \text{training conditions}) \mapsto \underbrace{\Pi_{S_0, \mathcal{E}}^\Lambda}_{\text{Machine Prior}}, \quad MesoMachine \sim \Pi_{S_0, \mathcal{E}}^\Lambda.$$

This law is the *Machine Prior*, or mechanistic inductive bias. Architecture does not determine one Machine; it changes the feasible CTM trade-offs and the probability with which training realizes them.

The theory links forward prediction, inverse causal recovery, and architecture-edit falsification. Typed reachability connects transport geometry to required sequential depth. A finite-cut versus addressable-memory law shows that independently queryable bindings must pay either exponentially growing robust distinguishability behind a fixed cut or growing exposed memory plus a selection interface. Quotient descent identifies when physical mechanisms implement composable computational operations. These results yield architecture-conditioned predictions for MLPs, CNNs, recurrent/state-space models, Transformers, and extensions. The CTM triad itself is not claimed as novel; the proposed contribution is to make its *learned causal organization* a predictive and recoverable dependent variable of architecture science.

A claim crosses the *Evidence Firewall* only when its measurement and intervention contracts license the corresponding CTM role. A recovered account earns the *MESO SEAL* only if it *Survives* unseen structure, *Edits* predictably, *Acts* causally, and *Links* compositionally. The framework is rejected if MesoGenome information adds no held-out predictive value about CTM mechanism fingerprints, failure interfaces, or edit responses beyond task, training, resource, and behavioral baselines.

Keywords: computational mesoscale; compute–transport–memory; MesoGenome; MesoGene; MesoMachine; Machine Prior; neural architecture; mechanistic interpretability; model behavior

1 Introduction

1.1 The dependent variable that architecture science is missing

Consider two models that answer the same associative-retrieval query with the same accuracy. One preserves source-local key–value records and later selects one by content. The other compresses the assignments through a recurrent cut and reconstructs the value at query time. Standard architecture comparison can report that both models express and learn the task. It cannot name the variable on which they differ, predict whether increasing load will cause collision or misaddressing, or state which architecture edit should move one implementation toward the other.

The missing dependent variable is the learned organization of *compute*, *transport*, and *memory*: which intermediate distinctions are transformed, where their consequences become available, what remains distinguishable for later use, how these roles coordinate, and where failure first enters. Behavioral equivalence underdetermines all of them. A task

can be solved by addressable records, compressed state, a scan, a shortcut, or an input-conditioned mixture. Figure 1 shows why the difference is scientifically consequential even before it changes accuracy.

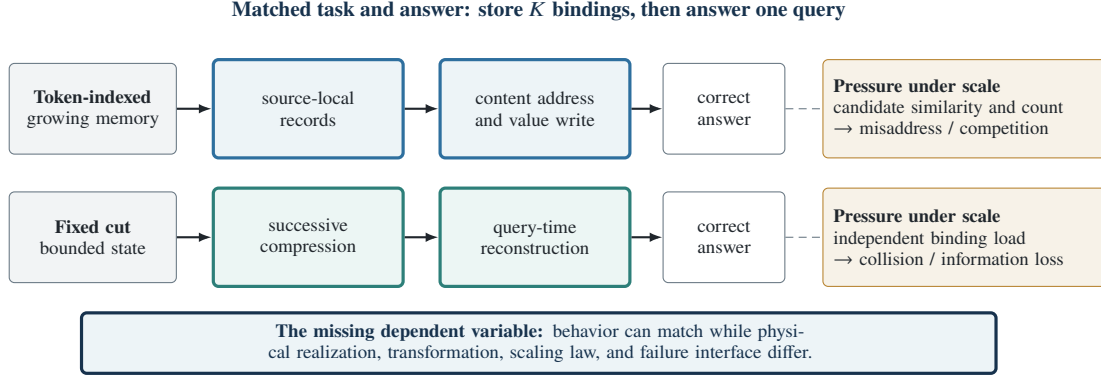


Figure 1: Same function, different CTM organization. A growing addressable memory and a fixed recurrent cut can produce the same answer at low load while using different compute, transport, and memory roles and predicting different failure scaling. This paper treats that learned organization as a dependent variable of architecture science. Solid arrows show the candidate execution order; dashed unheaded connectors associate each organization with its predicted scaling pressure and are not additional causal stages.

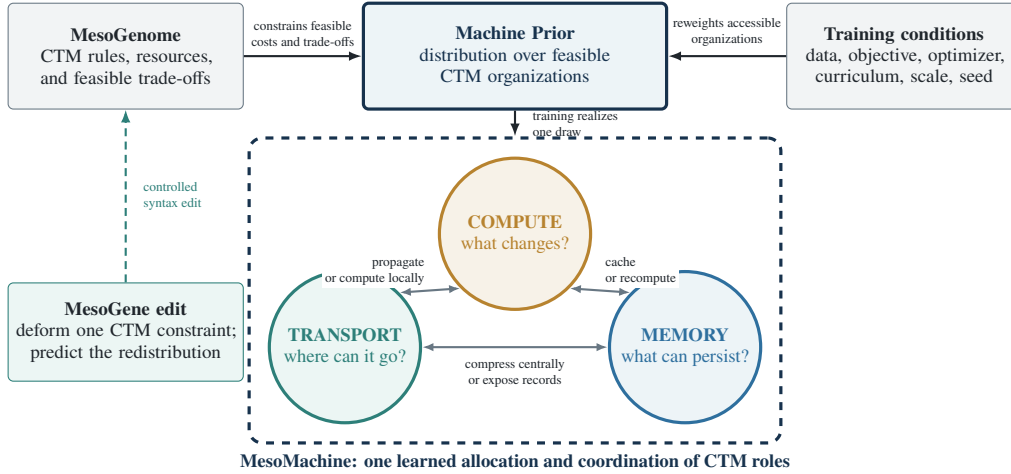


Figure 2: The generative semantics of the compute–transport–memory triangle. The MesoGenome constrains the feasible CTM cost surface; training conditions reweight it, together inducing a Machine Prior from which training realizes a MesoMachine. Inside the dashed Machine boundary, the three colored nodes are causal roles rather than modules, and the bidirectional edges are trade-offs, not causal arrows. Solid one-way arrows denote generative selection; the teal dashed arrow denotes a controlled MesoGene edit to the MesoGenome.

How to read the figures. Gold, teal, and blue denote evidence-licensed compute, transport, and memory roles; gray denotes context, method, or unclassified physical structure. Arrowheads are never decorative: their local label or caption states whether they mean execution, generative selection, recovery, or a controlled edit. Bidirectional edges denote a tested correspondence or trade-off, while dashed lines denote an alternative, boundary, association, or controlled contrast specified in that figure’s caption. No color or edge by itself establishes a causal role.

The broader research programme seeks a lawful passage from neural execution to organized model behavior. The present paper isolates its first target layer: *MESO*, the computational mesoscale. Its canonical proposition is compact:

Every neural architecture determines what may change, where computational consequences may move, and what may persist. Its MesoGenome sets these CTM conditions; training selects a MesoMachine from the accessible organizations. MESO is the science of predicting, discovering, and testing that learned Machine.

Architecture research already studies function classes, optimization, scaling, efficiency, and behavioral inductive bias. The proposal here is a second-order inductive bias. A fully specified causal syntax and set of training conditions induce a distribution over MesoMachines, evaluated at a declared equivalence level. We call this distribution the *Machine Prior* (the formal mechanistic inductive bias).

Theory claim. *Neural architectures bias not only which functions are learned, but which mechanisms realize them.* The MesoGenome constrains what can develop; training selects an internal organization; held-out causal experiments recover and test that organization.

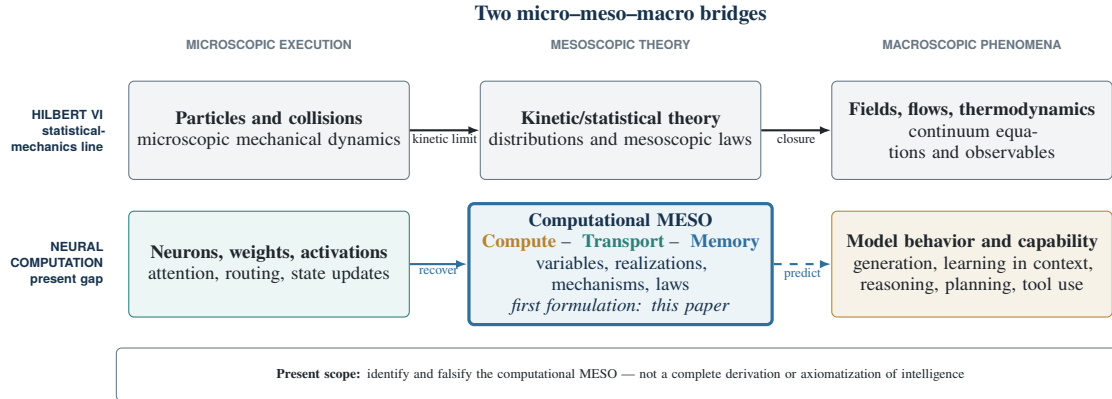
The statement is not that an architecture brand carries a fixed ontology. Rather, syntax changes the probability, stability, transport, and characteristic failure of candidate implementations. It predicts how controlled architecture edits should move mechanism distributions, not merely how a trained model can be redescribed after inspection.

1.2 Paper identity, scope, and contributions

This is a *foundational architecture-theory paper with a falsification protocol*: it introduces a scientific object, organizes its causal syntax through CTM, derives architecture-conditioned separations, and specifies how the theory can be estimated and rejected. The benchmark is not the paper’s identity; it is the confirmatory instrument required by the theory. The paper is neither a report of completed cross-architecture results nor a manifesto that treats future mathematics as present evidence. Its purpose is to make learned internal organization a predictive target of architecture science. The complete programme extends much further—through mechanism formation, operating-regime switching, cross-model transport, reconstruction, and architecture design—but this paper develops only the formal and experimental layer required to make those later questions scientific.

A Hilbertian programme for neural computation, bounded at the first bridge. Hilbert’s sixth problem called for axiomatic treatment of those physical sciences already receptive to mathematical formulation. One canonical line of that programme seeks controlled passages from microscopic mechanics, through kinetic or statistical descriptions, to continuum laws (Hilbert, 1902; Gorbunov, 2018). This paper invokes that *research architecture*, not the subject, maturity, or achieved status of the physical programme. The relevant lesson is that stable macroscopic regularities need not be read directly from individual particles: an intermediate vocabulary and its laws can be scientifically indispensable.

Neural systems present the same open shape. At the microscopic end, every parameter, activation, routed message, and state update is physically executable and increasingly observable. At the macroscopic end, trained models exhibit organized behavior—generation, retrieval, in-context adaptation, reasoning, planning, and tool use. Whatever philosophical threshold one adopts for the word *intelligence*, these structured capabilities are empirical phenomena that require explanation. What is missing is a disciplined causal level between them.



The parallel is deliberately asymmetric. In statistical mechanics, candidate micro-, meso-, and macro-theories can be named before the limiting relation is proved. In a trained neural system, the intermediate objects and laws are themselves unknown. The neural problem is therefore *inverse as well as axiomatic*: recover a stable intermediate theory from physical execution, then retain it only if it predicts held-out interventions, transport, composition, reconstruction, and failure. The computational MESO is not a particular neural layer or a convenient summary; it is a candidate causal level of description that compresses microscopic execution while preserving the distinctions needed to explain macroscopic behavior. This paper offers a first formulation and an evidence gate for that level, not a derivation of intelligence in full. Later closures would have to connect computational organization to structural semantics, grounding, agent–environment

Table 1: The Computational Mesoscale at the junction of neighboring research programmes. The rows are complementary, not claims of exclusive territory.

Programme	Primary question/output	Gap addressed here
Expressivity and complexity	Which functions or languages can the architecture realize under resource assumptions?	Does not determine which successful internal organization training selects
Behavioral inductive bias	Which functions, cues, or generalizations are learned more readily?	Usually treats internal mechanism as latent or secondary
Representation comparison	Which geometries, features, or alignments appear across models?	Similarity does not establish native transformations, composition, or failure interfaces
Mechanistic interpretability	Which internal variables and circuits causally contribute in a trained model?	Usually begins after architecture and does not estimate an architecture-conditioned mechanism distribution
Causal abstraction	Does a low-level system faithfully implement a proposed high-level causal model?	Often requires the high-level variables or alignment family to be supplied
Computational Mesoscale	Which CTM organizations are made relatively likely by syntax and training, recoverable under contracts, and shifted by architecture edits?	Unifies forward mechanism prediction, inverse recovery, and edit-based falsification

loops, and alignment under the same standards. Only then could the resulting stack become a common formal spine for model interpretability and alignment.

The paper makes five primary contributions.

1. It proposes CTM as a compact, falsifiable causal grammar of the computational mesoscale: compute changes distinctions, transport makes consequences available elsewhere, and memory preserves distinctions for later use.
2. It defines the *MesoGenome*, *MesoMachine*, and *Machine Prior*, joining architecture-conditioned forward prediction, inverse recovery, and MesoGene-edit causality.
3. It organizes typed reachability, robust cut capacity, quotient composition, and the finite-cut versus addressable-memory law as structural relations among CTM roles.
4. It derives falsifiable CTM profiles for four core architecture families and uses Transformer dereferencing as a worked audit of memory, address, transformation, publication, consumer, competition, and composition.
5. It supplies the Evidence Firewall and MESO SEAL required to state exactly what the evidence licenses and to accept or reject a recovered Machine.

1.3 Relation to neighboring programmes

The Computational Mesoscale does not replace expressivity, inductive-bias, representation-comparison, causal-abstraction, or mechanistic-interpretability research. It changes their junction. Classical architecture theory constrains possible functions and resources; empirical inductive-bias work observes behavior and representation geometry; mechanistic interpretability recovers local objects and processes; causal abstraction tests relations between low- and high-level models (Cohen and Welling, 2016; Ravanbakhsh, 2020; Geiger et al., 2023; Pearl, 2009). It asks whether architecture syntax predicts the *distribution* of recovered computational organizations and whether controlled MesoGene edits causally move that distribution.

The words compute, communication, and storage already organize computer architecture and I/O-complexity research; process theories likewise distinguish typed processes, wiring, and stateful feedback (Hong and Kung, 1981; Kissinger, 2012; Fong et al., 2015). The CTM triad is therefore not presented as a new systems taxonomy. The proposed increment is to treat the *learned, architecture-conditioned causal organization* of these roles as an endogenous variable: predict its distribution before inspection, recover it from a trained neural system, and perturb it through a declared architecture edit.

1.4 Roadmap

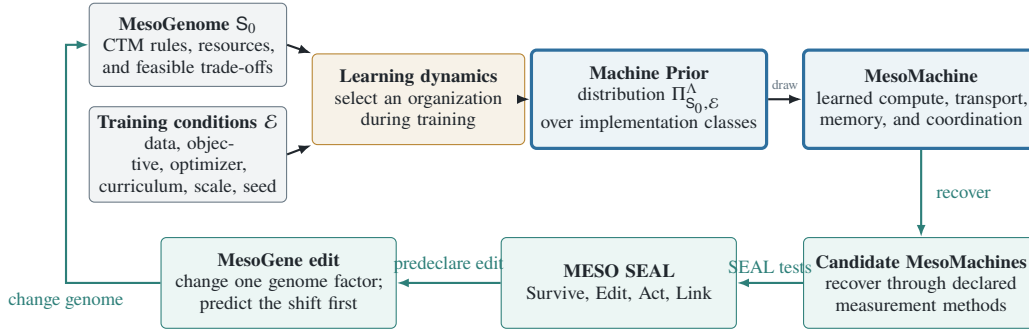


Figure 3: The generative and scientific loop in one view. A MesoGenome constrains feasible CTM organizations; ordinary training conditions and learning dynamics induce a Machine Prior and realize a MesoMachine. Measurement methods narrow the evidence-compatible candidates, and the MESO SEAL asks whether the recovered account can Survive, Edit, Act, and Link. The loop is scientific only when the MesoGene edit is declared before the new model is trained and inspected. Black arrows form the generative training path; teal arrows form the measurement–test–edit cycle and do not imply that evidence causes the original Machine.

2 The compute–transport–memory grammar

2.1 Three causal roles

CTM is a working causal decomposition of a learned Machine, not a claim that every architecture contains three visibly separated modules. A state or process may carry several roles, and role membership is licensed by interventions rather than by component name.

Definition 1 (Compute, transport, and memory roles). Within a declared experimental frame:

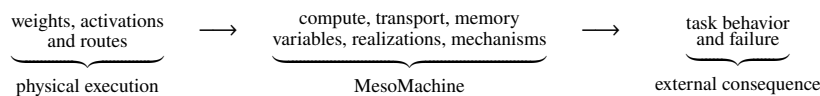
- Compute** a physical process has a compute role when it uniformly realizes a declared computational operation on the admitted realization family.
- Transport** a physical process has a transport role when an admitted computational distinction at one typed site becomes causally available at another while preserving the declared identity or operation.
- Memory** a physical state or process has a memory role when a distinction persists across a declared horizon and remains available to a later native computation.

The roles are relations, not mutually exclusive component labels.

The public vocabulary remains deliberately small. A *MesoGenome* is the full CTM syntax of an architecture; a *MesoGene* is one declared editable factor inside it. Training conditions and learning dynamics retain their ordinary names. The selected organization is a *MesoMachine*. Inside that Machine, a *computational variable* is an abstract distinction that participates in computation; a *physical realization* is a state, subspace, trajectory, record, or distributed configuration that instantiates it; a *computational operation* is an abstract map between variables; and an *execution mechanism* is a physical process that implements an operation uniformly over the declared realization family. The word *protocol* is reserved for coordination-heavy compositions such as writer–memory–address–reader systems. Operating regime, measurement method, mechanism fingerprint, and failure interface remain descriptive terms rather than additional branded objects. The *Evidence Firewall* prevents method output from being mistaken for the Machine itself.

Theory claim. Compute changes a distinction; transport makes its consequence available elsewhere; memory keeps it available for later use. Variables say what participates, physical realizations say how it exists, and execution mechanisms perform the role. A protocol coordinates several mechanisms across CTM.

The target sits between microscopic execution and macroscopic behavior:



Microscopic traces do not arrive pre-partitioned into computational objects, while task structure does not uniquely identify the model’s implementation. Interventions constrain the middle layer from below; task counterfactuals constrain it from above.

2.2 Notation and generative objects

The notation deliberately separates the external task, trained physical model, measurement method, evidence splits, and recovered implementation claim.

Table 2: Disjoint symbol families for task, model, method, data, and recovery objects.

Symbol	Kind	Meaning
$\mathcal{T}$	task specification	Externally defined abstract states, operations, observations, and task interventions
S_0	model syntax	Nominal typed CTM syntax: primitive operations, graph, order, sharing, symmetries, routing, memory interfaces, cuts, resources
θ	model parameter	A draw from stochastic training
$S_\theta^{\text{op}}(x, c)$	execution trace	Exact operational dependencies for one run, before empirical thresholding
$\mathfrak{F}$	experimental frame	Context scope, admissible interventions, native/readout outcomes, graph granularity
$M_{\theta, z}^{\text{meso}}(\mathfrak{F}, \varepsilon)$	MesoMachine	Stable variables, physical realizations, execution mechanisms, and evidence-licensed CTM roles for operating regime z at tolerance ε
$\mathfrak{Q}_m$	measurement method	Target, observables, interventions, alignment class, aggregation, scope, and cost
$\mathcal{D}_{\text{fit}}, \mathcal{D}_{\text{sel}}, \mathcal{D}_{\text{eval}}, \mathcal{D}_{\text{hold}}$	data	Disjoint method fitting, selection, evaluation, and structural holdout splits
$\widehat{\mathfrak{M}}_{\varepsilon, \delta}$	candidate set	MesoMachines not rejected at tolerance ε and uncertainty δ

2.2.1 Generation layer

Definition 2 (MesoGenome (nominal causal genome)). A model specification has a *MesoGenome*, formally its nominal CTM syntax,

$$S_0 = (\text{Ty}, \text{Op}_0, \mathcal{G}_0, <, \sim_{\text{par}}, G_{\text{exact}}, \rho_0, \text{Mem}_0, \text{Cut}_0, \text{Res}_0).$$

Ty lists physical state types; Op_0 lists the primitive operation and update families; $\mathcal{G}_0$ is the potential typed dependency graph; $<$ is execution order; $\sim_{\text{par}}$ is parameter sharing; G_{exact} contains only exact group actions; ρ_0 is the routing class; Mem_0 declares persistent state types, update/read/write interfaces, addressability, and nominal horizons; Cut_0 lists graph cuts at a declared node granularity; and Res_0 records finite-precision, bandwidth, memory, and compute assumptions. Approximate symmetries are not elements of G_{exact} ; they are metric claims with a defect and tolerance.

The CTM projection groups these ingredients by causal role:

$$\text{CTM}(S_0) = (\underbrace{C_0}_{\substack{\text{operations, update laws,} \\ \text{sharing and composition}}}, \underbrace{T_0}_{\substack{\text{paths, routing, position,} \\ \text{symmetry and bandwidth}}}, \underbrace{M_0}_{\substack{\text{persistence, addressability,} \\ \text{cuts, capacity and precision}}}).$$

This is not a disjoint factorization. Execution order, sharing, and resources can constrain more than one role. CTM is useful only if the resulting role profile predicts held-out mechanisms and edit responses better than the unstructured syntax list.

The genome metaphor marks a constraint, not a destiny: a MesoGenome limits and prices possible computational organizations without specifying one unique MesoMachine. A *MesoGene* is one preregistered, editable coordinate of this genome—for example parameter sharing, a memory cut, routing support, or a position rule. Architecture labels are not sufficient causal genomes. A “Transformer” specification must disclose mask, position system, normalization, residual organization, attention sparsity, cache semantics, precision, and any recurrence or expert routing. A “CNN” must disclose padding, stride, pooling, dilation, boundary convention, and readout. Cuts and resources are consequently frame- and granularity-indexed, not brand-level invariants.

Definition 3 (Training conditions and learning dynamics). Let $\mathcal{E}$ denote the *training conditions*: data distribution, objective, optimizer, curriculum, regularization, initialization law, scale, seed distribution, and implementation regime. The *learning dynamics* is the training map

$$\xi \sim p_{\Xi}, \quad \theta \sim \text{Train}(\mathbf{S}_0, \mathcal{E}, \xi).$$

This map describes how the MesoGenome and training conditions jointly produce a trained physical system; execution of that system realizes one or more scoped MesoMachines. No separate “compiler” object is assumed.

The forward problem predicts a distribution over operating regimes and MesoMachines, not a unique Machine:

$$p(z, \mathbf{M}_{\theta, z}^{\text{meso}} \mid \mathbf{S}_0, \mathcal{E}, x, c).$$

2.3 Operational execution and empirical effective syntax

Definition 4 (Operational execution syntax). $\mathbf{S}_{\theta}^{\text{op}}(x, c)$ is the fully resolved typed execution graph for one run: instantiated gates, selected experts, attention support before effect thresholding, residual and skip edges, normalization dependencies, shared randomness, caches, and sampler state. It is a property of the structural equations and runtime state, not of an interpretability method.

Operational support is not empirical causal importance. Attention weights, gate values, or nonzero Jacobian entries can coexist with zero relevant downstream effect. We therefore reserve

$$\widehat{\mathbf{S}}_{\theta}^{m, \mathfrak{F}, \mathfrak{C}_{\text{do}}, \varepsilon}(x, c)$$

for an *empirically recovered effective syntax*: edges or coalitions whose declared causal estimand exceeds tolerance ε under method m , frame $\mathfrak{F}$, and intervention contract $\mathfrak{C}_{\text{do}}$. This indexing removes the circular fiction that “causally active” edges are method-free model attributes.

2.4 Task, implementation, and recovery

Definition 5 (Task specification). The external task is

$$\mathcal{T} = (\mathcal{U}_{\mathcal{T}}, \mathcal{F}_{\mathcal{T}}, \mathcal{O}_{\mathcal{T}}, \mathcal{I}_{\mathcal{T}}, p_{\mathcal{T}}),$$

where $\mathcal{U}_{\mathcal{T}}$ is an abstract state set, $\mathcal{F}_{\mathcal{T}}$ a typed family of task operations, $\mathcal{O}_{\mathcal{T}}$ task observations, $\mathcal{I}_{\mathcal{T}}$ permitted task interventions, and $p_{\mathcal{T}}$ the generator. $\mathcal{T}$ is ground truth about the *task*; it is not ground truth about a trained model’s internal mechanism.

Definition 6 (Typed MesoMachine). For parameters θ , coherent operating regime z , frame $\mathfrak{F}$, and tolerance ε , a *MesoMachine* (formally, a typed implementation world) is

$$\mathbf{M}_{\theta, z}^{\text{meso}}(\mathfrak{F}, \varepsilon) = (\mathbf{U}_z, \mathbf{X}_z, \{\sim_{\tau, z}\}_{\tau \in \text{Ty}}, \text{Real}_z, \text{Exec}_z, \text{Role}_z, \text{Prot}_z, \text{Trans}_z, \Gamma_z).$$

$\mathbf{U}_z$ is a typed family of computational variables; $\mathbf{X}_z$ the scoped physical states and processes; $\sim_{\tau, z}$ the intervention-and-reader equivalence on physical realizations of type τ ; $\text{Real}_z \subseteq \mathbf{X}_z \times \mathbf{U}_z$ the realization relation (whose inverse images are *realization fibers* or *physical realization families*); Exec_z a typed family of execution mechanisms that preserve relevant equivalences;

$$\text{Role}_z \subseteq (\mathbf{X}_z \sqcup \text{Exec}_z) \times \{\mathbf{C}, \mathbf{T}, \mathbf{M}\}$$

the evidence-licensed CTM role relation; Prot_z the declared abstract/physical protocol-composition tests; Trans_z partial correspondences across contexts, positions, times, seeds, or models; and Γ_z records scope, approximation defects, resource costs, and unresolved alternatives. An internal transport role in Role_z concerns native movement within one Machine execution; Trans_z concerns tested correspondence across sites, regimes, or systems. They must not be identified.

Definition 7 (Evidence-compatible candidate set). After method fitting and selection are frozen, recovery returns

$$\widehat{\mathfrak{M}}_{\varepsilon, \delta} = \text{Recover}(\mathcal{D}_{\text{eval}}, \mathcal{D}_{\text{hold}} \mid \mathbf{S}_0, \mathcal{E}, \mathfrak{Q}_m, \mathfrak{F}, \mathfrak{C}_{\text{do}}, \mathfrak{C}_{\text{id}}),$$

the set or posterior of MesoMachines not rejected at error tolerance ε and uncertainty level δ . Recovery is evaluated by held-out interventional prediction, compression under a declared complexity class, typed composition, transport, and constructive tests. Metaphysical uniqueness is not an output.

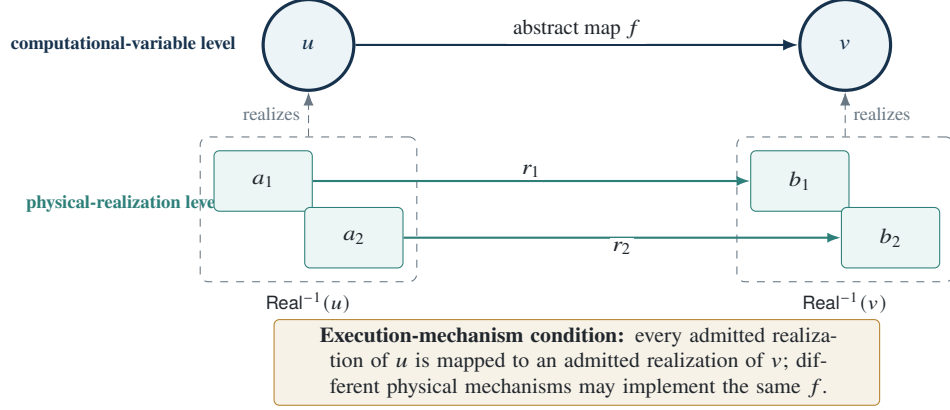


Figure 4: Anatomy of the compute role inside a MesoMachine. Computational variables are equivalence classes supported by one or more physical realizations. A physical process becomes an execution mechanism only when it acts uniformly across the declared realization family and descends to the abstract map. Transport and memory require additional availability and persistence tests; they are not inferred from this diagram alone.

The three objects are connected but not identified:

$$\mathcal{T} \xrightarrow[\text{training}]{S_0, \mathcal{E}, \xi} \mathcal{M}_{\theta, z}^{\text{meso}} \xrightarrow[\text{measurement}]{\mathcal{Q}_m, \mathcal{F}, \mathcal{C}_{\text{do}}} \mathcal{D}_{\text{eval}} \xrightarrow[\text{recovery}]{\mathcal{C}_{\text{id}}} \widehat{\mathcal{M}}_{\mathcal{E}, \delta}.$$

A task operation can have multiple execution mechanisms; a trained model can use a shortcut not isomorphic to the task decomposition; and several candidate MesoMachines can predict the same observed evidence.

2.5 Operating regimes and cross-input coherence

Allowing an arbitrary Machine for every input makes the theory unfalsifiable. We therefore introduce a *operating regime*

$$z \sim p_{\psi}(z \mid x, c, \theta), \quad z \in \mathcal{Z},$$

where $\mathcal{Z}$ and its granularity are fixed before final evaluation by one of three declared rules: a planted label, a preregistered finite family, or a complexity-penalized partition selected on $\mathcal{D}_{\text{sel}}$. Each z has a stable MesoMachine on a scope $\Omega_z \subseteq \mathcal{X} \times \mathcal{C}$. The coherent family of these scoped Machines and tested transports is simply the model's family of operating regimes. New regimes cannot be created after inspecting $\mathcal{D}_{\text{hold}}$.

For a declared context arrow $\gamma : c \rightarrow c'$, transport is a partial correspondence

$$\mathbf{X}_{z, c} \xleftarrow{\pi_c} K_{\gamma} \xrightarrow{\pi_{c'}} \mathbf{X}_{z', c'},$$

not necessarily an invertible conjugacy. Identity and cocycle defects are evaluated on the common domain:

$$\Delta_{\text{id}}(\gamma) = d(\text{Trans}_{\text{id}_c}, \text{id}), \quad \Delta_{\text{cyc}}(\gamma_2, \gamma_1) = d(\text{Trans}_{\gamma_2} \circ \text{Trans}_{\gamma_1}, \text{Trans}_{\gamma_2 \circ \gamma_1}).$$

Birth and death mean failure to extend a computational-variable correspondence above a preregistered coverage threshold; split and merge mean a change in the cardinality of non-gauge equivalence classes under a fixed correspondence rule. They are empirical outcomes, not post-hoc explanatory escape hatches.

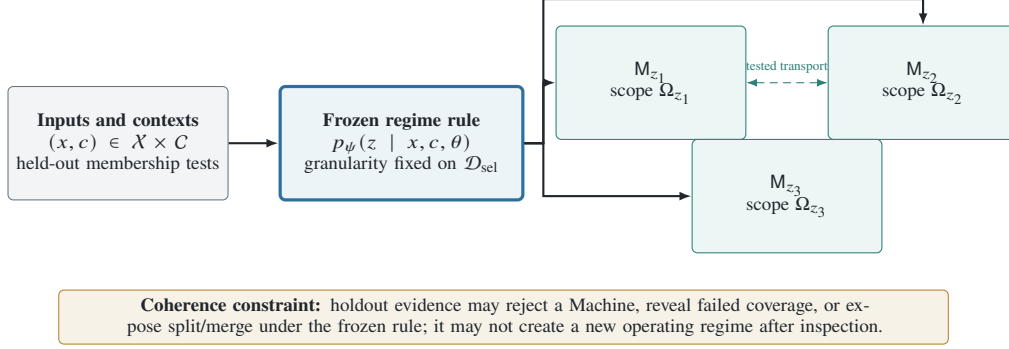


Figure 5: A coherent family of operating regimes. Input- and context-dependent switching is permitted only through a preregistered or selection-frozen regime rule. Each regime has a stable scope, and cross-context transport is a separately tested partial correspondence. This prevents “one post-hoc Machine per input” from making the theory vacuous. Solid arrows are regime assignments under the frozen rule; the teal dashed two-way arrow is a separately tested correspondence and does not imply that the two Machines cause one another.

3 Machine Prior: the forward theory

Classical inductive bias is usually observed through functions learned, sample efficiency, or generalization. The present framework asks for the induced distribution over *implementations* among models that meet a declared behavioral criterion.

Definition 8 (Machine equivalence level). Let Λ specify the claim type, gauge, scope, reader/intervention family, and tolerance used to compare MesoMachines. Write

$$M \equiv_\Lambda M'$$

when the two Machines satisfy the declared equivalence criterion. Examples include behavioral, abstract-algorithm, variable/relation, execution mechanism-family, physical-coordinate, and actionability-profile equivalence. These are distinct relations, not steps in a total order.

Definition 9 (Machine Prior). Fix architecture syntax S_0 , training conditions $\mathcal{E}$, comparison level Λ , and evaluation scope. Training and coherent mechanism selection induce

$$\xi \sim p_\Xi, \quad \theta \sim \text{Train}(S_0, \mathcal{E}, \xi), \quad z \sim p(z \mid \theta, x, c),$$

and hence a probability law

$$\Pi_{S_0, \mathcal{E}}^\Lambda := \mathcal{L}\left([M_{\theta, z}^{\text{meso}}]_\Lambda\right)$$

over MesoMachine equivalence classes. This law is the *Machine Prior* of the specified architecture–training system on the declared scope; formally, it is the system’s mechanistic inductive bias.

Definition 10 (CTM cost profile and feasible set). For a task family $\mathcal{T}$, declared resource model, and behaviorally admissible MesoMachine M , let

$$\mathbf{K}_{S_0}(M; \mathcal{T}) = (K_C, K_T, K_M, K_{\text{coord}})$$

be a vector of compute, transport, memory, and cross-role coordination costs. Each coordinate is separately operationalized; no universal weighted sum is assumed. The architecture-relative feasible set is

$$\mathcal{F}_{S_0, \mathcal{T}} = \{\mathbf{K}_{S_0}(M; \mathcal{T}) : M \text{ realizes } \mathcal{T} \text{ within the declared behavior region}\}.$$

Its Pareto frontier represents implementation trade-offs under the declared resource and equivalence conventions, not a universal ordering of Machines.

Architectures do not determine Machines; they constrain and price CTM organizations. The MesoGenome shapes the feasible set; training conditions and optimization determine which accessible regions receive probability mass. A MesoGene edit can deform the feasible frontier, move the Machine Prior, or do both. This account assumes neither a Gibbs law nor a common currency for CTM costs. A measurement method changes the evidence likelihood and recoverability of Π ; it does not become an argument of the generative law. The experimental frame and Λ remain necessary because there is no unscoped notion of Machine identity.

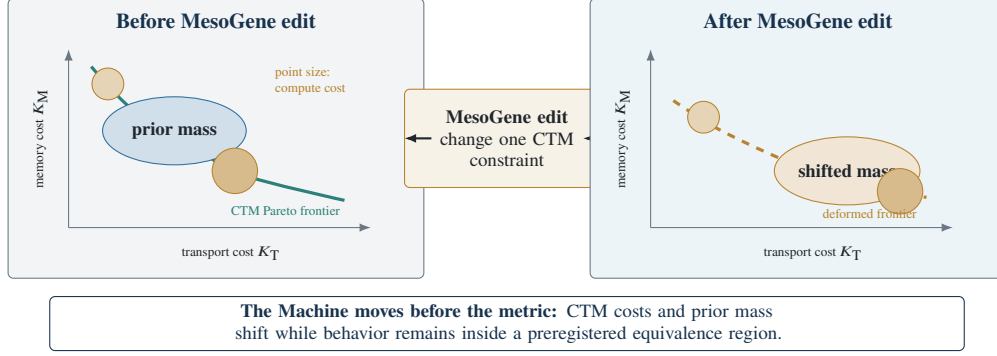


Figure 6: A schematic CTM trade-off surface, not reported data. Axes represent separately declared costs rather than a universal scalar; point size illustrates a third coordinate. A controlled MesoGene edit can deform the feasible Pareto frontier and move Machine-Prior mass before behavior separates.

3.1 Three linked scientific problems

Forward prediction. Predict $\Pi_{S_0, \mathcal{E}}^\Lambda$ or a preregistered mechanism fingerprint

$$\sigma(M) = (\sigma_C, \sigma_T, \sigma_M, \sigma_\times, \sigma_{\text{fail}}, \sigma_{\text{act}})$$

from syntax and training before inspecting the final holdout internals. The first three coordinates are role-specific profiles:

σ_C = (operation interchange, compute depth/cost, withheld composition),

σ_T = (reachability, routing adaptivity, transport defect, competition),

σ_M = (persistence, addressability, robust capacity, interference).

$\sigma_\times$ records coordination terms such as cache–recompute, route–transform, and store–select interactions. These are estimand families, not entries in a universal scalar score.

Inverse recovery. Given a trained physical system and evidence produced under declared measurement and experimental contracts, recover a set or posterior of evidence-compatible MesoMachines over Machine-equivalence classes:

$$\text{Recover} : (\mathfrak{P}_\theta, \mathfrak{Q}, \mathfrak{F}, \mathcal{D}) \longrightarrow \mathbb{P}([M]_\Lambda \mid \mathcal{D}).$$

Architecture-edit causality. Under a coupled training design, change one syntax component and estimate the effect on mechanism fingerprints:

$$\Delta_{S_0 \rightarrow S'_0} \mathbb{E}[\sigma(M)] = \mathbb{E}_{\Pi_{S'_0, \mathcal{E}}^\Lambda}[\sigma] - \mathbb{E}_{\Pi_{S_0, \mathcal{E}}^\Lambda}[\sigma].$$

When exact pairing is unavailable, the estimand is defined over matched training populations with explicit assumptions rather than called an individual-model causal effect.

Prediction 11 (The Machine moves before the metric). For some operationally active MesoGene edits, the distribution of preregistered mechanism fingerprints will move detectably while task performance remains inside a declared equivalence region. The strongest version predicts the direction of that movement before holdout internal inspection. Failure to find such predictive surplus across matched tasks weakens the claim that architecture contributes anything beyond behavioral inductive bias.

3.2 Core predictive quantities

The framework has a clear empirical increment only if architecture information predicts mechanisms beyond task and training descriptors. Let $\mathcal{M}_0$ predict a mechanism fingerprint Y from $\mathcal{T}, \mathcal{E}$ and behavioral summaries, while $\mathcal{M}_1$ additionally receives S_0 . The primary architecture-aware advantage is the held-out proper-score difference

$$\Delta_{\text{arch}} = \mathbb{E}_{\text{hold}}[S(\mathcal{M}_1, Y) - S(\mathcal{M}_0, Y)].$$

Other population quantities are the entropy or dispersion of Π^Λ , the checkpoint at which a Machine class becomes predictively stable, and the distribution of typed failure interfaces. These quantities require a declared candidate class or fingerprint; no universal scalar “mechanism distance” is assumed.

Theory claim. The first payoff is a separation, not a stereotype. For K independently queryable bindings, a fixed robust cut must distinguish q^K joint configurations, while K addressable memory cells can store them separately and answer by selection. The architectures need not choose these implementations, but if they do, their dominant pressure moves from *compression* to *address selection*. [Theorem 19](#) states the result formally.

4 The inverse problem: measurement contracts and the Evidence Firewall

4.1 The measurement method and independent data

Definition 12 (Measurement method). An interpretability *measurement method* m (formally, a method syntax) is

$$\mathfrak{Q}_m = (\text{Target, Obs, Do, Align, Agg, Scope, Cost}),$$

recording the claim it targets, observables, admissible interventions, alignment class, aggregation rule, scope, and supervision/search/compute budget. A measurement method controls what can be seen and tested; it is not a transparent window onto the Machine.

Fitted methods use four disjoint roles:

$$\mathcal{D}_{\text{fit}} \rightarrow \text{fit probes/SAEs/alignments}, \quad \mathcal{D}_{\text{sel}} \rightarrow \text{select hyperparameters and candidate Machines},$$

$$\mathcal{D}_{\text{eval}} \rightarrow \text{estimate preregistered claims}, \quad \mathcal{D}_{\text{hold}} \rightarrow \text{test unseen interventions, compositions, and transports}.$$

Donor selection, prompt search, automated-agent trajectories, and stopping decisions belong to fitting or selection. Reusing them for final evidence requires selective-inference correction and is otherwise prohibited.

4.2 Intervention and identifiability contracts

Definition 13 (Intervention contract). A causal claim is indexed by

$$\mathfrak{C}_{\text{do}} = (\text{Estimand, Outcome, Treatment, Reference, Donor, Coupling,} \\ \text{Support, Reader, Interactions, Estimator, } \varepsilon, \delta, \text{Correction, Stopping}).$$

The contract states the causal estimand; outcome and native reader; treatment/reference states; donor-recipient coupling; admissible support and off-manifold diagnostics; interaction order; estimator; equivalence or superiority tolerance ε ; uncertainty δ ; multiple-testing correction; and stopping rule.

Definition 14 (Identifiability contract). $\mathfrak{C}_{\text{id}}$ specifies the candidate-Machine class, alignment family, allowed gauges, intervention and reader families inherited from $\mathfrak{C}_{\text{do}}$, complexity prior, scope, alternative hypotheses, and an observational/interventional indistinguishability threshold. It answers when two candidate Machines can be separated. It does not redefine a single causal effect.

The contracts compose as follows: $\mathfrak{C}_{\text{do}}$ defines the evidence likelihood for each candidate Machine; $\mathfrak{C}_{\text{id}}$ defines the equivalence relation and posterior/model-selection problem over the candidate MesoMachines.

4.3 The Evidence Firewall

Firewall principle. A measurement is evidence about the Machine; it is not the Machine. An interpretability output is licensed by a declared measurement method and experimental contract. It is not, by itself, a computational variable, native variable, causal mechanism, or privileged coordinate system of the model.

The firewall is not skepticism about measurement. It is a type discipline: a probe result, sparse code, attribution score, interchange effect, reconstruction, and transplant answer different questions. Promotion to a stronger claim requires the missing experiment rather than a change in vocabulary.

Table 4: Default licensed conclusions. A method may support a stronger conclusion only by supplying the missing evidence.

Evidence	Default license	Does not by itself license
Held-out probe	Decodability in a specified class	Native use, causality, natural coordinates
Sparse dictionary	Sparse reconstruction in a chosen basis	Causal-variable, Jacobian, or pathway sparsity
Attribution/mask	A declared localization estimand	Necessity, sufficiency, or interaction completeness
Natural-donor interchange	Counterfactual use within support	Global identity or unique mechanism
Ablation/rescue	Scoped necessity/sufficiency	Portability or compositional closure
Reconstruction / transplant	Constructive actionability	Metaphysical uniqueness
Withheld composition	Typed compositional prediction	Higher context laws outside the test scope

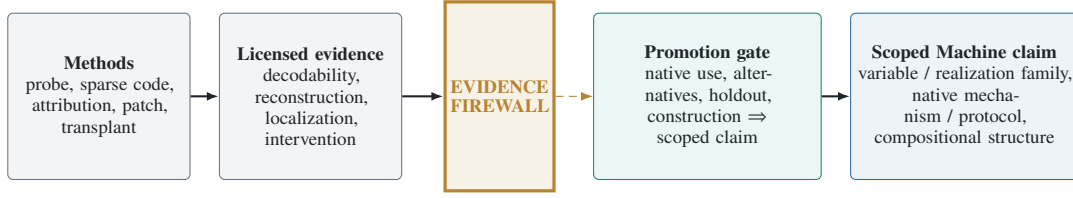


Figure 7: The Evidence Firewall. A measurement method generates evidence; its declared contract states what that evidence permits, and the firewall blocks every stronger default inference. The dashed crossing is licensed only by the named promotion tests; the final solid arrow records the resulting scoped claim. The diagram is an evidence-licensing order, not a model-execution graph.

Table 3: The CTM evidence matrix. Color or architectural location never assigns a role; each role requires a distinct counterfactual test. Coordination claims require factorial or coalition experiments across roles.

Role	Candidate sensors	Decisive intervention	Forbidden inference
Compute	Local input–output map, Jacobian/finite differences, operation probes, component attribution	Interchange inputs while holding the physical memory realization and route fixed; test withheld operation and typed composition	“Activated” or “predictive” $\Rightarrow$ the component natively performs the operation
Transport	Attention/message patterns, causal edge effects, routing gates, donor–recipient alignment	Move or swap a natural donor while holding abstract value and downstream transform fixed; test delivery and use	High attention, edge weight, or route frequency $\Rightarrow$ transported content or causal contribution
Memory	Pulse–chase decodability, persistence curves, state/cache occupancy, reader access	Write, erase, delay, and rescue the nominated physical memory state while holding generators and later demand fixed	Persistent decodability $\Rightarrow$ native storage, necessity, or addressability
CTM coordination	Cross-role timing, coalition effects, cache/recompute and route/transform coupling	Factorial edits and ordered coalitions; compare direct with composed protocols on structural holdout	Individually causal parts $\Rightarrow$ a compositional MesoMachine

Evidence profiles, not a scalar ladder. Interpretability targets are neither a total order nor a mathematically defined lattice. We report a profile

$$\mathbf{e} = (e_{\text{read}}, e_{\text{dec}}, e_{\text{loc}}, e_{\text{use}}, e_{\text{nec}}, e_{\text{suf}}, e_{\text{act}}, e_{\text{trans}}, e_{\text{comp}}, e_{\text{id}})$$

whose coordinates are separately defined estimands with uncertainty. A componentwise partial order is used only when two profiles share frame, scale, and tolerance. No “license gap” is formed by subtracting incomparable coordinates. Readability, actionability, reconstruction, and transport can be incomparable.

Four sparsities. The evidence synthesis motivates four distinct sparsities:

$$S_{\text{act}}, \quad S_{\text{var}}, \quad S_{\text{jac}}, \quad S_{\text{path}},$$

for activations, realized causal variables, Jacobian/computation, and physical pathways. Their units and failure modes differ:

S_{act} — **activation sparsity.**

Few coordinates are nonzero in a chosen activation basis. It licenses an economy of description in that basis, not a claim about causal variables.

S_{var} — **realized-variable sparsity.**

A small set of interventionally distinguished variables accounts for the tested counterfactuals. It depends on the intervention/readout frame and does not imply a sparse physical path.

S_{jac} — **computational sparsity.**

Few input-state directions exert locally non-negligible effects on declared downstream variables, for a stated Jacobian or finite-difference scale. It can be dense even when activations are sparse.

S_{path} — **pathway sparsity.**

A small physical coalition of edges, heads, neurons, or modules carries the tested causal effect under an interaction-aware estimand. Redundancy and self-repair can make a sparse native protocol look dense under isolated ablation.

None implies another without additional assumptions. Sparse autoencoders are therefore candidate coordinate systems and sensors, not privileged MesoMachines (Cunningham et al., 2023; Gao et al., 2024a; Farnik et al., 2025; Arora et al., 2026; Mueller et al., 2025; Wu et al., 2025b). The distinction is experimentally productive: a method should be evaluated against the sparsity it actually optimizes, not rewarded for a different sparsity by verbal transfer.

4.4 Multi-axis evidence coding

Single letters are replaced by

$$\mathbf{g}(s, k) = (I_{\text{int}}, I_{\text{do}}, I_{\text{match}}, I_{\text{gt}}, R, T, V),$$

for source s and claim k . The first four coordinates are binary: internal measurement, intervention, architecture-matched alternatives, and mechanism ground truth. $R \in \{0, 1, 2, 3\}$ records replication/task/model breadth; $T \in \{0, 1, 2, 3\}$ records theory-assumption strength, with higher values meaning stronger assumptions; $V \in \{\text{preprint, conference, journal, book}\}$ records publication status. Mixed labels such as “A/B” are disallowed. The supplement records each cited claim at this resolution.

5 Structural relations of compute, transport, and memory

The CTM triad becomes useful only when its edges make predictions. This section formalizes three such edges. Compute requires transport whenever an operation depends on a physically remote state. Transport across a bottleneck constrains what memory can preserve. Compute and memory trade persistence against recomputation. We then state the quotient condition under which these physical processes descend to composable mesoscale maps. Reachability is standard in structural causal and computation graphs; predictive quotients are related to automata minimization, predictive-state representations, and bisimulation; quotient descent is elementary (Pearl, 2009; Nerode, 1958; Singh et al., 2012; Ferns et al., 2004; Geiger et al., 2023). The contribution is the CTM organization of these ingredients and the edit-level predictions it supports, not a claim to have invented communication, storage, or computation.

5.1 Compute–transport: typed reachability

Lemma 15 (Communication-path necessity). *Let a deterministic typed acyclic computation graph have structural equations*

$$v = F_v(\text{pa}(v))$$

for every node v . Add explicit nodes or hyperedges for residual/skip paths, global pooling or normalization, cache state, cross-sample statistics, and shared randomness. If no directed path connects intervention target u to outcome node v , then any two admissible interventions on u with all other exogenous variables coupled identically produce the same v . In a stochastic graph, the corresponding interventional distributions are equal under the same coupling and structural assumptions.

Proof. Order nodes topologically. Nodes outside the descendants of u have identical parents and exogenous variables under both interventions, hence identical values by induction. Since v is not a descendant of u , it is invariant. The stochastic statement follows by conditioning on, or identically coupling, exogenous variables. $\square$

Corollary 16 (Local path lower bound). *If each layer permits edges of metric radius at most r_ℓ , an effect across distance d requires a path satisfying $\sum_{\ell=1}^L r_\ell \geq d$. This is only a path-existence lower bound. It does not imply that a detectable or usable effect grows smoothly or monotonically with d .*

5.2 Transport–memory: cut factorization and robust capacity

Let a history h cross a declared cut through state $c(h) \in \mathcal{C}$, and let $\Sigma(h)$ denote its vector of future interventional outcome distributions under a specified experiment family. The predictive equivalence is

$$h \sim_\Sigma h' \iff \Sigma(h) = \Sigma(h').$$

Proposition 17 (Predictive cut factorization). *If $\Sigma(h)$ depends on h only through $c(h)$, then the quotient map $q_\Sigma : h \mapsto [h]_{\sim_\Sigma}$ factors through the cut image: there exists a unique map $\bar{q}$ on $c(\mathcal{H})$ such that*

$$q_\Sigma = \bar{q} \circ c.$$

The proposition does not assert that the learned cut state is minimal, disentangled, or sufficient for outcomes outside the declared experiment family.

Proof. If $c(h) = c(h')$, the assumption gives $\Sigma(h) = \Sigma(h')$, so $h \sim_\Sigma h'$. Therefore $\bar{q}(c(h)) = [h]_{\sim_\Sigma}$ is well-defined and unique on the cut image. $\square$

Finite width alone does not imply finite information capacity for real-valued states. A capacity claim requires noise, precision, robustness, or a mutual-information constraint.

Proposition 18 (Robust cut capacity). *Suppose q^K task configurations must be distinguished after a cut. Let their cut codes lie in a metric set (C_{reach}, d) and require pairwise margin greater than 2γ for a downstream reader to be robust to perturbations of radius γ . Then*

$$q^K \leq N_{\text{pack}}(C_{\text{reach}}, d, 2\gamma),$$

where N_{pack} is the packing number. If C_{reach} lies in a d -dimensional Euclidean ball of radius R , then

$$K \log q \leq d \log \left(1 + \frac{2R}{\gamma} \right).$$

Alternatively, if decoding error is at most α for a uniformly distributed configuration Y , Fano’s inequality implies

$$(1 - \alpha)K \log q - h_2(\alpha) \leq I(Y; C).$$

Proof. Robustly distinguishable codes form a 2γ -packing. The Euclidean bound is the standard volumetric packing bound. The information statement is Fano’s inequality rearranged. $\square$

The empirical question is consequently not whether a vector has “finite width,” but which robust packing or information bound applies under declared precision, noise, margin, reader class, and error tolerance.

Theorem 19 (Finite-cut versus addressable-memory law). *Let a task present K independently variable bindings $Y = (Y_1, \dots, Y_K) \in [q]^K$, followed after a memory boundary by a query $I \in [K]$ whose required answer is Y_I .*

1. Suppose every influence of Y on the answer crosses a single state $C(Y) \in \mathcal{C}$, and correctness must be robust to perturbations of radius γ at that cut. Then the q^K cut codes must form a 2γ -packing:

$$q^K \leq N_{\text{pack}}(C_{\text{reach}}, d, 2\gamma).$$

In particular, any architecture family whose robust cut packing number is bounded independently of K must eventually incur nonzero worst-case error.

2. Suppose instead that the memory boundary exposes an extensive family of addressable memory cells $C_K = (C_1, \dots, C_K)$, each cell has q robustly distinguishable states, and the query node has a permitted path to every cell. Then an exact implementation exists: write Y_i into C_i , use I to select C_I , and emit its value. The required number of cells grows linearly with K while the nominal query-to-memory path length need not grow with K .

Proof. For the first statement, if $Y \neq Y'$, then some coordinate i differs. Because query $I = i$ must return different answers, the two cut codes must be distinguishable by the downstream reader. Robustness to radius γ requires their perturbation balls to be disjoint, hence their distance exceeds 2γ . All q^K assignments therefore form a packing. For the second statement, choose q robust codes for each memory cell, store coordinate Y_i in cell i , and let the query select and decode cell I . This construction uses K cells and the declared direct path. $\square$

Corollary 20 (Relocation of failure pressure). *For the task family in Theorem 19, increasing K places exponential distinguishability pressure on a fixed robust cut. An extensive addressable memory removes that particular fixed-cut lower bound by growing memory with K , but introduces a different empirical interface: selection among records and delivery to the downstream consumer. Thus an architecture edit can relocate the dominant failure pressure without changing low-load accuracy.*

Theory claim. The transport–memory exchange rate. Independent bindings must be paid for somewhere. A fixed robust cut pays through exponentially growing distinguishability; an extensive addressable memory pays through additional records, transport bandwidth, and a selection interface. This compression-to-selection shift does not say which implementation training chooses. It predicts how a CTM trade-off relocates resource pressure and the failure interface.

Remark 21. The theorem is a structural separation, not a claim that every recurrent model uses a minimal compressed state or every Transformer learns source-local records. A recurrent architecture may enlarge or externalize its state; an attention architecture may compress history. The prediction concerns the resource scaling and failure interfaces of the two implementation classes.

5.3 Compute–memory: persistence and recomputation

Memory can preserve a reusable intermediate result, while computation can reconstruct that result from retained generators. Which strategy is cheaper is architecture-, task-, and regime-relative: recomputation needs a path back to the generators and sufficient compute; persistence needs a stable physical memory realization and a later reader. The two strategies can coexist or be selected conditionally.

Prediction 22 (Persistence–recomputation response). Consider a task with a repeatedly consumed intermediate variable u , a candidate persistent physical realization of u , and retained generating inputs from which u can be recomputed. If the persistent realization is selectively erased while the generating inputs and downstream demand remain available, a recomputation-based Machine should increase later compute depth, activation, or latency while preserving the abstract output within tolerance. A storage-dependent Machine should instead lose or corrupt the output unless an alternative memory realization or backup path activates. Conversely, adding an operational cache should reduce repeated compute only when later consumers use that cache. The prediction is inapplicable when the generators are unavailable or the edit changes task semantics.

This is a directional edit law rather than a universal time–space theorem. It separates three possibilities that behavioral accuracy alone conflates: native persistence, native recomputation, and intervention-induced repair.

5.4 Quotient descent and CTM composition

Proposition 23 (Typed quotient descent). *Let $r : X \rightarrow Y$ and $s : Y \rightarrow Z$ be typed physical processes, with equivalence relations $\sim_X, \sim_Y, \sim_Z$. If*

$$x \sim_X x' \Rightarrow r(x) \sim_Y r(x'), \quad y \sim_Y y' \Rightarrow s(y) \sim_Z s(y'),$$

then unique maps

$$\bar{r} : X/\sim_X \rightarrow Y/\sim_Y, \quad \bar{s} : Y/\sim_Y \rightarrow Z/\sim_Z$$

are induced by $\bar{r}([x]) = [r(x)]$ and $\bar{s}([y]) = [s(y)]$, and

$$\overline{s \circ r} = \bar{s} \circ \bar{r}.$$

Proof. The preservation assumptions make both maps independent of representative. For every $[x]$,

$$\overline{s \circ r}([x]) = [s(r(x))] = \bar{s}([r(x)]) = (\bar{s} \circ \bar{r})([x]).$$

$\square$

Corollary 24 (Symmetry action on a quotient). *If a group acts equivariantly on typed states, interventions, and outcomes, future-intervention equivalence is preserved and the group acts on the quotient by $g \cdot [x] = [g \cdot x]$.*

Table 5: Four core architecture portraits in CTM space. Each cell describes a relative implementation cost or resource, not a compulsory learned ontology. The final column turns the portrait into a causal test.

Family	Compute	Transport	Memory	Predicted pressure → decisive MesoGene edit
Dense / tied MLP	Dense static transforms; tying can reuse an operator	Short global paths but no canonical slot transport unless imposed	Layer state is transient; no native sequence-growing record set	Unseen-slot exchangeability → add parameter tying while retaining reach
CNN / G-CNN	Shared local operator and hierarchical aggregation	Geometric local propagation; exact group action depends on boundaries and readout	Distributed feature fields, residual persistence, optional multiscale bottlenecks	Distance and boundary effects → dilation/global edge or boundary edit
RNN / SSM	Time-shared state update; selective variants condition the update	Sequential recurrent channel through a mandatory cut	Bounded recurrent state with gate- or mode-structured retention	Collision, overwrite, or decay → enlarge/bypass the cut or alter retention
Transformer	OV/MLP transforms and cross-layer protocol composition	QK/content-dependent selection over short potential paths	Token-indexed residual/KV records, plus optional compression	Address competition and interface failure → edit attention, position, or cache syntax

Remark 25. [Corollary 24](#) supplies an action on quotient classes. It does not identify orbit members as one object, guarantee low physical transport defect, or guarantee that a chosen method recovers the action. Those are separate empirical claims.

For approximate symmetry, let each transition have equivariance defect at most η_ℓ and Lipschitz constant L_ℓ . A length- n composition obeys the telescoping bound

$$\eta_{1:n} \leq \sum_{k=1}^n \eta_k \prod_{j=k+1}^n L_j.$$

Approximate identity is represented by a pseudometric or cover, not by an ε -threshold relation falsely assumed transitive.

6 Architecture portraits in CTM space

The structural results eliminate impossible mechanisms; they do not select a unique remaining Machine. An architecture portrait therefore describes a feasible CTM region and a conditional prior, never an architectural personality. Every claim is typed as a hard consequence, an evidence-supported tendency, or an open conjecture. The probabilistic target is

$$\Pi_{S_0, \varepsilon}^\Lambda,$$

with scope, competing explanations, and a MesoGene edit that can move the distribution. [Table 5](#) is the visual index; the subsections justify each row and specify its decisive edit. Evidence cited here is claim-scoped; evidence axes and publication status appear in the integrated appendices and source files.

6.1 Untied and tied MLP baselines

CTM portrait. **Compute** is a dense, trained transformation and can be reused across slots only when tying, symmetry, or training makes that reuse economical. **Transport** has short potential path length but fixed support and no canonical correspondence between coordinates. **Memory** is ordinarily transient layer state rather than a dedicated, sequence-growing address space. An untied dense MLP has short nominal path length and fixed potential support, but no canonical coordinate transport. A tied or equivariant MLP adds sharing or a group action. A static token mixer has a fixed mixing matrix, yet nonlinear gates can still produce strongly input-dependent *effect magnitudes*; fixed support must not be equated with low causal adaptivity ([Ravanbakhsh, 2020](#); [Wang et al., 2022](#); [Yu et al., 2021](#)).

Conditional prior. Cheap global dependence and costly exchangeability are the baseline expectation. The same abstract operation at two coordinates need not share an execution mechanism unless tying, data symmetry, or optimization creates one.

Prediction 26 (MLP transport prior). On an exchangeable-slot task, an untied MLP should have a larger held-out slot-transport defect than a parameter-tied control at matched task performance. The claim concerns transport, not a unique “holistic ontology.” It is rejected if the difference lies within a preregistered equivalence margin.

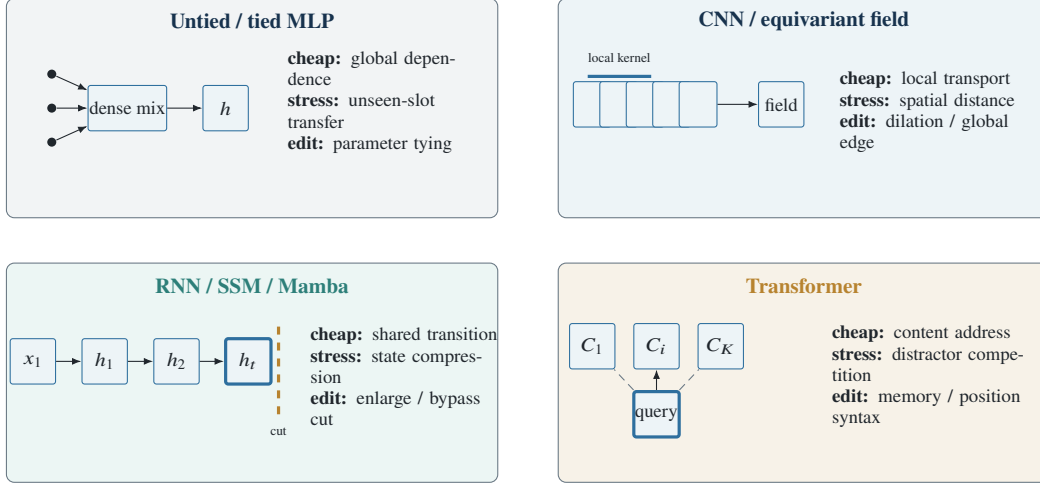


Figure 8: Four architecture families and their conditional Machine Priors. The cartoons do not assign a unique ontology to an architecture. They show where each syntax makes a candidate organization relatively direct, where scaling pressure is expected to enter, and which controlled edit can test the account. Arrows inside a panel show nominal information-flow directions allowed by that syntax, not recovered causal edges; the dashed Transformer lines are unselected candidate addresses.

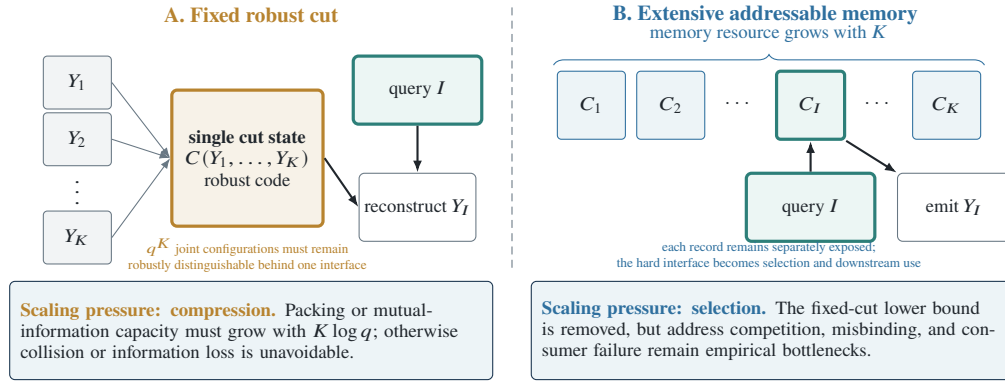


Figure 9: The finite-cut versus addressable-memory law. The same retrieval behavior can be implemented through two memory organizations with different resource scaling and different failure interfaces. [Theorem 19](#) makes the left-hand pressure formal and gives the right-hand construction. It does not assume which organization training selects. Arrows show the direction in which records and queries enter the answer computation; the top brace denotes resource scaling, not information flow.

Decisive edit. Introduce tying while keeping potential reach fixed. A reduction in cross-slot execution mechanism defect before, or without, an accuracy change supports the syntax account. Dynamic mixing is separate: it tests conditional source selection, not transport by itself.

6.2 Convolutional networks

CTM portrait. **Compute** is organized around shared local operators and their depth-wise composition. **Transport** follows spatial adjacency and any declared group action. **Memory** appears as distributed feature fields, residual persistence, or multiscale bottleneck/skip states rather than one separate memory module. CNNs combine local communication and spatial parameter sharing. Exact translation equivariance depends on boundary, stride, pooling, sampling, and readout; zero padding can leak absolute position ([Cohen and Welling, 2016](#); [Islam et al., 2021](#); [Luo et al., 2016](#)). Group-equivariant CNNs make a larger exact action explicit, while residual and U-Net paths alter reach and cuts.

Conditional prior. When task, data, boundary, intervention, and reader preserve a group action, position-indexed causal fields and reusable local execution mechanisms should be cheaper than coordinate-specific rediscovery. [Corollary 24](#) only guarantees an action on a quotient; treating orbit-related realizers as the same variable additionally requires

native-reader interchange and a declared identity criterion.

Prediction 27 (CNN transport and propagation). In successful models on a task that requires a relation across controlled distance, symmetry-preserving CNNs should show: (i) lower shift-transport defect than untied locally connected controls; and (ii) a positive relationship between required distance and the earliest *detectable native causal effect*, subject to censoring and effective receptive field. Neither statement follows from path existence alone.

Boundaries. Shape/texture preference is not a syntax invariant: data and augmentation can reverse or suppress it (Geirhos et al., 2019; Hermann et al., 2020; Burgert et al., 2025). Hierarchical composition is a prior, not a theorem; pooling, shortcuts, and global readouts can bypass it.

Decisive edits. Compare circular, valid, and zero-padded variants; add dilation or a global edge while preserving the task; and compare tied with locally connected untied filters. The mechanism prediction is directional: padding should increase coordinate information; dilation/global edges should reduce causal mediation depth; tying should reduce transport defect.

6.3 Recurrent and state-space models

CTM portrait. **Compute** is a shared recurrent update, linear mode evolution, or an input-conditioned selective update. **Transport** is sequential and must cross the recurrent interface. **Memory** is the evolving state, with retention set by gates, modes, precision, and state size. RNNs and SSMs route all prefix influence through a recurrent state at every time. [Proposition 17](#) makes this a factorization statement, not a claim that the state is minimal or predictive. Time-shared parameters do not guarantee time-invariant execution mechanisms: gates, boundary state, input distribution, and saturation change effective maps (Hochreiter and Schmidhuber, 1997; Downey et al., 2017; Venkatraman et al., 2017).

RNN prior. Task-sufficient cut states and shared transition protocols are relatively cheap. Under increasing concurrent binding load, robust finite-resource models should exhibit mergers, overwrite, or forgetting when the required distinguishability exceeds the cut’s packing/information budget. Fixed dimension alone is not the premise.

Linear SSM prior. For

$$h_t = Ah_{t-1} + Bx_t, \quad y_t = Ch_t + Dx_t,$$

past inputs enter as mode-filtered superpositions. Retention lifetimes and terminal reconstruction are therefore natural candidates, but their realization depends on spectrum, discretization, precision, parameterization, and optimization (Gu et al., 2021; Wang and Li, 2024; Jelassi et al., 2024; Merrill et al., 2024).

Selective SSM precision. Canonical Mamba keeps a learned continuous-time A and skip D while making the discretization step Δ_t and input/output projections B_t, C_t input-dependent:

$$\bar{A}_t = \exp(\Delta_t A), \quad h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t, \quad y_t = C_t h_t + D x_t.$$

A short convolution precedes the selective scan (Gu and Dao, 2023). Selectivity changes retention and readout but does not create unbounded robust addressable memory.

Prediction 28 (Cut versus address). On matched associative-recall tasks, recurrent models should more often place decisive information in the evolving or terminal recurrent state, while token-indexed Transformers should more often preserve source-local records. This is a scoped distributional prediction, not an SSM ontology. Evidence from AR/ATR supports it only for the tested architectures and settings (Arora et al., 2025).

Transformer and Mamba may nonetheless implement analogous induction algorithms while differing in temporal deployment; SAE similarity alone does not establish causal identity (Wang et al., 2024). State size, short convolution, selectivity, and initialization are causal edits that should move the mechanism distribution (Trockman et al., 2024).

6.4 Transformers

CTM portrait. **Compute** is carried by OV/MLP transformations and by their cross-layer composition. **Transport** is implemented through content-dependent QK selection and residual delivery. **Memory** can be realized by token-indexed residual/KV records or by learned compression. This is a role assignment, not a clean module partition: a head, token

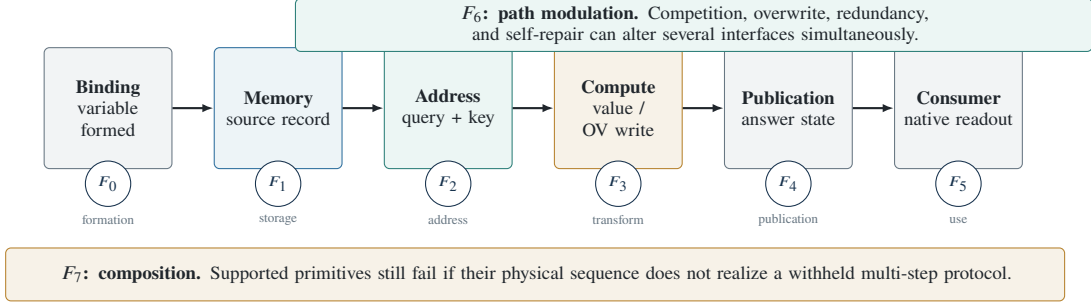


Figure 10: A stage-resolved Transformer CTM audit. Blue marks a memory role, teal a transport/address role, and gold a compute/transform role. Information can be present at one interface and fail at a later one; competition and repair can alter several stages simultaneously. The labels are causal hypotheses to be distinguished by interventions, not mutually exclusive post-hoc error names. The left-to-right arrows specify the hypothesized execution order to be tested; they do not claim that each stage occupies one component or one layer.

state, or residual path may participate in several roles under different contexts. Transformers provide token-indexed residual states that can serve as physical realizations or memory records, short nominal path length under full attention, content-dependent QK selection, and OV-dependent written content (Vaswani et al., 2017; Elhage et al., 2021). Attention weights specify neither causal edge effects nor downstream use. The empirically effective graph depends on values, output projection, competing heads, residual paths, and native readers. KV cache is an inference implementation of growing token memory, distinct from the abstract full-sequence training graph.

Relative prior. Content-addressed protocols are cheap, not compulsory. A useful candidate unit is typed:

$$\text{writer} \rightarrow \text{memory record} \rightarrow \text{address/query} \rightarrow \text{transform/write} \rightarrow \text{consumer}.$$

Induction circuits and variable-binding studies provide concrete multi-component instances (Olsson et al., 2022; Wu et al., 2025a; Merullo et al., 2024). The same architecture can also learn scans, compressed states, convolution-like maps, heuristics, or input-conditioned mixtures (Li et al., 2025; Cordonnier et al., 2019; Li et al., 2021; Pîslar et al., 2025).

6.4.1 Worked CTM audit: symbolic dereferencing

Consider a controlled program

$$[A := v_1] [B := v_2] \cdots [Q := B] [\text{answer} = ?],$$

with optional chains such as $Q := C$ and $C := B$. Surface names, values, positions, and irrelevant assignments are independently randomized. The *task specification* contains bindings and typed dereference operations. It does not specify how a trained Transformer implements them.

Competing MesoMachines. At least three candidates must be retained during discovery.

1. **Address protocol** $\mathcal{H}_{\text{addr}}$. Assignment sites preserve source-local records; later queries construct a role/address; QK-like selection locates a record; an OV-like map writes a value into a publication state; a downstream consumer uses it.
2. **Compressed-state or scan protocol** $\mathcal{H}_{\text{scan}}$. Assignments are successively folded into a task-sufficient state. The query is answered from that state rather than by source-local dereference.
3. **Heuristic or mixture** $\mathcal{H}_{\text{mix}}$. Absolute position, recency, name frequency, or a small candidate set supports a shortcut on part of the distribution, with a different protocol activated elsewhere.

The candidate class also includes refinements and observationally equivalent variants. Its granularity and coding language are frozen on $\mathcal{D}_{\text{set}}$; otherwise every error can be redescribed as a new Machine.

Table 6: Protocol audit for the Transformer dereferencing example. Rows are candidate causal interfaces, not mutually exclusive labels or an assumed temporal hierarchy.

Interface	Candidate evidence	Discriminating intervention	What a positive result licenses
Formation	Abstract value or binding can be decoded at a nominated site	Natural-donor interchange across matched assignments	Scoped realization at the tested site; not downstream use
Memory/storage	Information persists under pulse-chase and matched perturbations	Replace or erase the nominated record while holding query and reader fixed	Storage dependence within the declared support
Address/query	Query and key variables predict source selection	Swap address variables while preserving values and positions	Address-mediated selection, conditional on downstream write
Transform/write	Selected source affects the proposed publication state	Intervene on value/OV map with source selection held fixed	Typed source-to-publication transformation
Publication	A recoverable answer state is written before final readout	Patch publication state while leaving earlier records unchanged	Availability to a declared downstream consumer
Consumer/readout	Native reader uses the publication state	Disable and rescue reader with matched publication donors	Scoped consumption/actionability
Competition/repair	Effects change with candidate similarity, count, or co-ablation	Factorial distractors; Harsanyi/ANOVA coalitions; conditional co-ablation	Interaction, redundancy, or self-repair in the tested coalition
Composition	Primitive dereferences are individually supported	Withhold and test typed two-step dereference $g \circ f$	Compositional prediction, not a universal algebra

Separation of discovery and test. Probes, sparse coordinates, attention patterns, and attribution are used on $\mathcal{D}_{\text{fit}}$ only to nominate storage, address, write, and consumer candidates. Hyperparameters, candidate partitions, and interaction coalitions are selected on $\mathcal{D}_{\text{sel}}$. Native-donor interventions and stage-specific rescues estimate declared effects on $\mathcal{D}_{\text{eval}}$. The decisive tests use $\mathcal{D}_{\text{hold}}$: unseen names and values, unseen absolute positions, longer contexts, factorially controlled distractor count and similarity, and withheld dereference compositions. No representation discovered on the holdout may enter the claim.

Evidence calibration. Induction circuits and controlled variable-binding systems support the possibility of $\mathcal{H}_{\text{addr}}$, while state-tracking experiments show that a Transformer can select different algorithms under curriculum, initialization, or task variation (Olsson et al., 2022; Wu et al., 2025a; Li et al., 2025; Zhong et al., 2023). Long-context studies establish position and utilization failures, but do not by themselves identify which interface failed (Liu et al., 2024; Gao et al., 2024b; Du et al., 2025). Distributed intervention and self-repair results motivate coalition tests rather than singleton rankings (Cheng and Zhang, 2026; Gong et al., 2026; Vaidyanathan et al., 2026). The worked example therefore demonstrates the framework; it does not establish that natural-language Transformers universally use an address ontology.

Retrieval geometry. Full attention removes a distance-growing *reachability lower bound*; it does not remove position, length, normalization, competition, or reader effects. The testable estimand is not an unspecified function $P(\text{error}) = f(\cdot)$, but a preregistered factorial contrast over absolute position, relative distance, context length, candidate count, candidate similarity, and position syntax. Long-context behavior depends on absolute position and length even under successful retrieval (Liu et al., 2024; Du et al., 2025; Gao et al., 2024b).

Prediction 29 (Reach versus retrieval). After matching absolute position, context length, and distractor statistics, increasing relative source distance should not produce the path-length scaling required by a local CNN. Increasing candidate similarity or count should increase address competition. Superiority or equivalence is assessed with preregistered contrasts, not qualitative “low/high” labels.

Storage/use dissociation. Decodable target information can coexist with incorrect output (Gao et al., 2024b; Karim et al., 2026; Basu et al., 2026). This licenses a dissociation, not the conclusion that editing a query or reader will repair the model. Repair requires stage-specific intervention.

Protocol interactions. Single-component patching can mis-rank conjunctive, redundant, or self-repairing systems (Vaidyanathan et al., 2026; Cheng and Zhang, 2026; Gong et al., 2026). Group intervention is expected to dominate

isolated intervention only under a declared interaction structure. We use Harsanyi dividends,

$$\Delta_H(S) = \sum_{T \subseteq S} (-1)^{|S|-|T|} \Delta(T),$$

or a preregistered causal interaction estimand, rather than the uncalibrated difference between a coalition and the sum of singleton effects.

Position syntax. Virtual-variable transport depends on the joint position syntax

$$\text{Pos} = (\text{encoding, mask, normalization, BOS/sinks, residuals, training range}).$$

Absolute, relative, rotary, randomized, and no-explicit-encoding systems have different behavioral length-generalization profiles (Haviv et al., 2022; Kazemnejad et al., 2023; Ruoss et al., 2023; Sinha et al., 2022; Ruscio et al., 2026). These results do not by themselves establish causal object identity across positions. That requires a natural-donor transport test with the same abstract counterfactual at source and destination.

Machine distributions. Mechanism diversity must be reported at a preregistered comparison level: behavior, abstract algorithm, variable structure, execution-mechanism family, or physical coordinate. Entropy is meaningful only after fixing this equivalence and reporting sensitivity to alternate partitions. Curriculum, initialization, checkpoint, scale, and input can change the selected Machine (Zhong et al., 2023; Li et al., 2025; Yin and Steinhardt, 2025; Pîslar et al., 2025).

Failure-interface profile. We use a multi-label causal graph, not a mutually exclusive “earliest failed interface.” The profile records formation, memory/storage, address/query, transform/write, publication, consumer/readout, competition/overwrite, and composition failures. Several may co-occur or be masked by repair. A posterior over them is reported only if a generative model and likelihood are specified; otherwise the output is a vector of intervention results.

Prediction 30 (Transformer protocol prior). At matched performance, Transformer errors on binding/retrieval tasks should be more recoverable by address-, competition-, publication-, or consumer-targeted interventions than recurrent errors whose relevant information has already collapsed before query time. The comparison is distributional and task-scoped; formation and storage failures remain possible in Transformers.

Decisive MesoGene edits. Compare full with local attention at fixed position syntax; change absolute, relative, rotary, randomized, and no-explicit-position systems; compress or evict token memory; and add a recurrent bottleneck while retaining comparable parameterized transforms. The predicted quantities are not merely accuracy and attention shape, but source-local storage mass, execution mechanism transport, competition effects, protocol composition, and the failure-interface distribution.

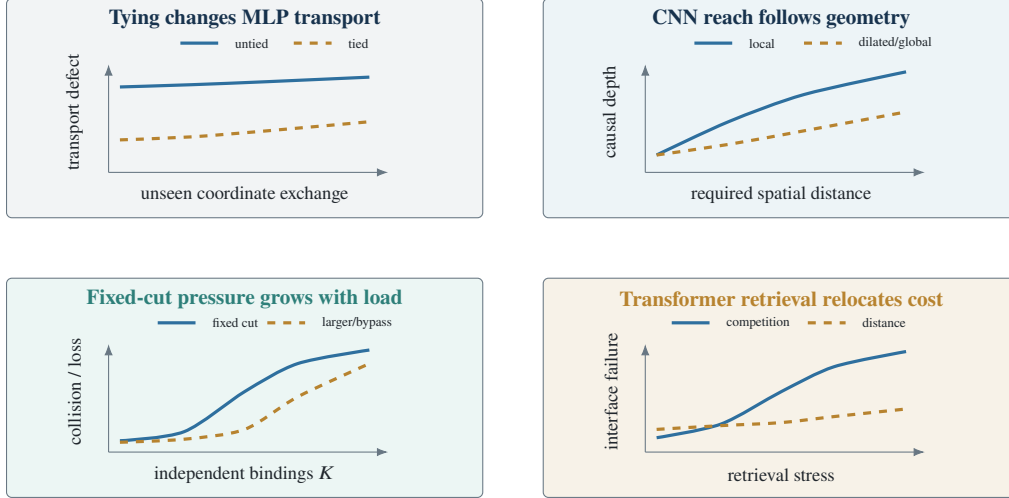


Figure 11: Predicted qualitative mechanism fingerprints, not reported data. Each panel pairs a stress variable with a MesoGene edit that should move the curve: tying should improve MLP transport; dilation or global edges should reduce CNN mediation depth; additional or bypassed state should delay fixed-cut collision; and Transformer retrieval should be more sensitive to candidate competition than to controlled source distance. The decisive evidence is a directional shift, not the architecture label.

How to read the atlas. The four panels are not phenomenological portraits of architecture classes. They are *response-shape predictions*. A syntax claim earns explanatory credit only when an intervention that changes the named structural resource changes the corresponding curve in the preregistered direction. Thus the scientific object is not a static label such as “CNN-like” or “Transformer-like,” but a paired contrast

(stress response before edit, stress response after edit).

This packaging separates a merely compatible story from a causal theory of Machine Prior.

Why the fingerprints form one theory. The panels share a common grammar: identify the physical realization of task-relevant distinctions, locate the interface through which those distinctions must pass, increase a stress variable, and edit the syntax that prices the candidate implementation. Tying changes the cost of transport; dilation changes the cost of reach; state expansion changes the cost of compression; and attention or memory edits change the cost of selection. The curves differ because the same abstract burden is paid at different physical interfaces.

6.5 Extension syntaxes in the same language

The core contrast is not limited to four architecture families. Three extensions expose different causal geometries while preserving the same question: what is cheap to carry, where is the mandatory interface, what failure should scale first, and which MesoGene edit should move it? Detailed technical boundaries and evidence appear in [Appendix C](#); the mechanistic predictions belong to the main theory.

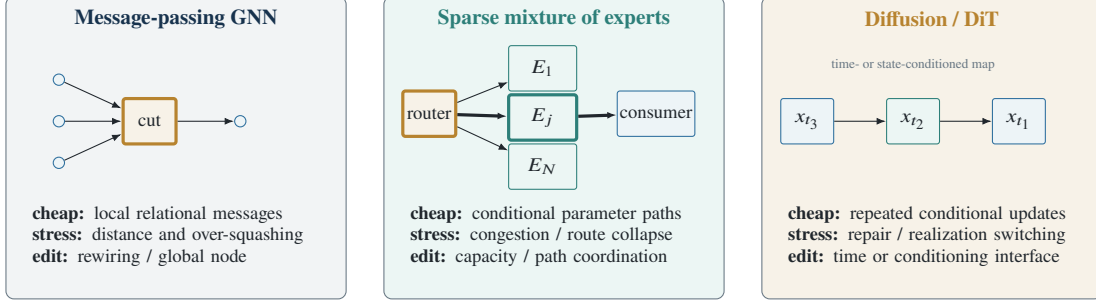


Figure 12: The extension panorama. GNNs, sparse MoE systems, and diffusion/DiT models instantiate graph-local aggregation, discrete conditional paths, and repeated state- or time-conditioned updates. The panels state conditional Machine Priors and decisive edits, not established architecture ontologies. Within-panel arrows show native execution direction only—message aggregation, router-to-expert-to-consumer flow, or reverse-time update—and do not imply relations between architecture families.

Graph-local aggregation. For a message-passing GNN, the native physical realization is indexed by nodes, edges, and neighborhoods, while graph distance bounds the earliest possible mediation. The distinctive pressure is not a generic “long-range problem”: information can be lost through over-squashing at a graph cut, homogenized through oversmoothing, or remain unidentified because the message-passing class cannot separate the relevant structures. Rewiring, a global node, or an enlarged cut must therefore move the corresponding mechanism fingerprint before it earns credit as an explanation.

Conditionally routed parameter paths. For sparse mixtures of experts, expert identity is not the computational object. The candidate mechanism is the complete router–expert–path–consumer protocol, including capacity constraints and token dropping. A semantic histogram over experts is weak evidence unless the route predicts held-out transformation and downstream use. Decisive edits change router coordination, active capacity, or expert paths while measuring congestion, path stability, collateral effects, and function-preserving transport.

Repeated time-conditioned Machines. For diffusion and DiT systems, a shared parameter block can implement different effective maps at different noise or flow times. The object of explanation is therefore a family $M_{\ell,t,x,c}$, not a static layer atlas. Physical-realization switching, repair, amplification, and trajectory bifurcation are distinct hypotheses. Direct transport from t_1 to t_3 must be compared with transport composed through t_2 ; edits to time conditioning, backbone connectivity, or the conditioning interface should move the predicted trajectory-level failure.

7 A calibrated measurement language

A compact formula is not an operational metric until it specifies the estimand, estimator, null, tolerance, and power target. Table 7 gives the core cards. All estimators are cross-fitted: candidate discovery uses $\mathcal{D}_{\text{fit}}$ and $\mathcal{D}_{\text{sel}}$, while estimates use $\mathcal{D}_{\text{eval}}$ and structural generalization uses $\mathcal{D}_{\text{hold}}$.

The measurement output is organized into three role profiles and their coordination terms:

$$\begin{aligned}\widehat{\Sigma}_C &= (\text{operation interchange, compute depth/cost, } \Delta_o), \\ \widehat{\Sigma}_T &= (\text{reachability, routing effects, transport defect, competition}), \\ \widehat{\Sigma}_M &= (\text{persistence, cut sufficiency, robust capacity, addressability, interference}).\end{aligned}$$

Cross-role measurements include cache–recompute response, route–transform interaction, storage–selection coupling, and ordered protocol composition. Reporting keeps the profiles separate: a model can have strong memory evidence but weak native use, or short transport paths but high selection error.

Table 7: Core estimand cards. “Natural” means sampled from the preregistered support rather than optimized post hoc.

Target	Estimand	Estimator and null	Calibration
Symmetry transport	Difference in native outcome under transformed natural donor versus matched source counterfactual, minimized over declared gauge	Cross-fitted paired effect; null is equality to coordinate-shuffled or untied control	Equivalence margin ε_{sym} ; group sampling law; familywise correction; target power $1 - \beta$
Propagation	Slope/finite differences of first layer with native effect above ε_{eff} as task distance changes	Interval-censored regression because “not detected” is censored; null slope set in advance	Distance grid, effect threshold, power at each layer, and multiplicity correction
Routing	Four-vector: support turnover, normalized edge-effect drift, written-content drift, downstream-use drift	Natural-donor interventions; compare inputs matched on scale and geometry; attention weights are not effects	Separate equivalence margins; report vector, never a scalar routing score
Cut sufficiency	Expected proper-score regret for predicting future intervention outcomes from cut state in a fixed hypothesis class	Cross-fitted probabilistic decoder versus history-aware and label-shuffled controls	Decoder class, future-experiment law, train/eval split, confidence interval, and complexity penalty
Robust capacity	Packing lower bound or mutual information needed for K simultaneous variables	Adversarial/noise-radius test and Fano-compatible decoder bound	Precision, noise, margin, allowed error, reader class, and state scaling
Execution-mechanism transport	Change in typed counterfactual effect under a partial physical-realization correspondence	Span/correspondence matching; null is donor shuffle or gauge-only alignment	Coverage threshold, non-invertible cases, equivalence margin, and uncovered mass
Composition	Native-outcome loss of physical $r_g \circ r_f$ on withheld task composite $g \circ f$	Decode/realize through native outcome map $\omega_Z : Z \rightarrow \mathcal{O}$; compare direct composite, shuffled order, and task-only baseline	Withheld composite family, type checks, equivalence/superiority margin, and sample size
Multiplicity	Lower bound on discovered non-gauge realization classes within a bounded search family	Bayesian or MDL model comparison after interchange-equivalence testing	Search class, gauge group, posterior threshold, and sensitivity analysis
Mechanism distribution	Predictive distribution over preregistered mechanism classes or an MDL partition	Hierarchical model across seeds/checkpoints/inputs; task-only versus syntax-conditioned log predictive density	Fixed equivalence level, held-out architectures/tasks, partition sensitivity, uncertainty

7.1 Typed composition

The physical and abstract outputs cannot be compared in one untyped loss. For execution mechanisms

$$r_f : X \rightarrow Y, \quad r_g : Y \rightarrow Z,$$

let $\omega_Z : Z \rightarrow \mathcal{O}_Z$ be a native outcome map and let $v_{g \circ f}(u) \in \mathcal{O}_Z$ be the task-specified counterfactual outcome. The held-out composition defect is

$$\Delta_o = \mathbb{E}_{(a,u) \sim p_{\text{hold}}} \ell_{\mathcal{O}_Z}(\omega_Z(r_g(r_f(a))), v_{g \circ f}(u)).$$

The composite $g \circ f$, donor family, and metric are withheld from recovery. A low primitive error with high Δ_o is evidence for a parts atlas without compositional closure.

7.2 Interaction and self-repair

Coalition effects are not reduced to singleton rankings. For component set N , estimate the set function $\Delta(S)$ for preregistered coalitions and decompose it with Harsanyi dividends or a specified causal ANOVA. Positive high-order terms indicate conjunctive dependence; negative terms can indicate redundancy or repair. Conditional co-ablation tests components in the background where they are predicted to become necessary. Order is treated explicitly for writer-reader protocols.

7.3 No premature scalarization

Concurrent-variable stress produces a vector:

$$\mathbf{I}(K) = (\Delta_{\text{object}}(K), \Delta_{\text{exec}}(K), \Delta_o(K), \text{FDR}_{\text{object}}(K), \text{coverage}(K)).$$

Arbitrary weighted sums are not used. Likewise, actionability is a vector of correction rate, collateral damage, robustness, transfer, and compute, rather than a subtraction from a localization score.

8 Evidence, counterevidence, and synthesis principles

The literature does not yet constitute a unified theory of architecture-conditioned internal computation. It provides fragments: matched architecture comparisons, mechanism case studies, formal constraints, representation analyses, intervention failures, and architecture edits. This section asks which parts of the Computational Mesoscale account are already well supported, which are scoped, and which remain conjectural. It is a selective synthesis of cited primary and formal sources, frozen July 27, 2026, not an exhaustive or PRISMA-complete review. The machine-readable supplement codes the key claim-bearing subset used in the synthesis, including the exact claim, evidence axes, publication status, and limitation; the broader bibliography also contains contextual and formal sources not represented as separate atlas rows.

8.1 Replicated or mature findings

1. **Behavior underdetermines representation and mechanism.** Probe controls, representation non-identifiability, and causal-abstraction work show that predictive fit alone does not establish native causal variables (Hewitt and Liang, 2019; Ravichander et al., 2021; Locatello et al., 2019; Geiger et al., 2023; Méloux et al., 2025; Sutter et al., 2025).
2. **Architecture components change structural affordances.** Locality/sharing, group equivariance, recurrent cuts, attention selection, and SSM state dynamics have clear formal or architectural consequences (Cohen and Welling, 2016; Ravanbakhsh, 2020; Vaswani et al., 2017; Gu et al., 2021; Gu and Dao, 2023).
3. **Position and context length affect Transformer retrieval.** Multiple studies show position-dependent long-context performance and position-system-dependent extrapolation (Liu et al., 2024; Haviv et al., 2022; Sinha et al., 2022; Kazemnejad et al., 2023; Ruoss et al., 2023; Du et al., 2025).
4. **Sparse reconstruction is not a privileged ontology.** Probe/benchmark comparisons, random-network results, neuron-basis circuits, and Jacobian sparsification separate activation sparsity from causal and computational sparsity (Heap et al., 2025; Arora et al., 2026; Farnik et al., 2025; Mueller et al., 2025; Wu et al., 2025b).

8.2 Strong but scoped causal findings

1. Matched AR/ATR experiments report induction-style source-local storage in tested Transformers/Based models and terminal-state computation in most tested SSMs; the claim is scoped to those tasks and models (Arora et al., 2025).
2. Transformer variable binding can use token-indexed addressable memory and specialized routing heads in controlled symbolic programs (Wu et al., 2025a).
3. Curriculum can shift a Transformer between state-tracking algorithms, and initialization/hyperparameters can produce different algorithmic solutions (Li et al., 2025; Zhong et al., 2023).
4. Natural-donor and multi-position interventions show that decodability can dissociate from use and that causal units can be distributed (Gao et al., 2024b; Cheng and Zhang, 2026).

8.3 Methodological cautions

1. Saliency, attention, probes, trained lenses, stitching, and sparse dictionaries estimate different targets and can pass misleading proxy tests (Adebayo et al., 2018; Jain and Wallace, 2019; Wiegreffe and Pinter, 2019; Belrose et al., 2023; Csiszarik et al., 2021; Chanin, 2026).
2. Single-mediator patching can contain interaction terms, and self-repair can hide backup pathways (Vaidyanathan et al., 2026; Gong et al., 2026).
3. Formal identifiability requires assumptions and does not ensure finite-sample or optimization recovery (Khemakhem et al., 2020; Schölkopf et al., 2021; Gamella et al., 2025).
4. Near-perfect decodability need not yield a reliable low-collateral causal handle (Basu et al., 2026).

8.4 Proposed synthesis hypotheses

The combined record supports several claims at different epistemic levels. They are not pooled into “established laws.” Table 8 separates structural consequences, methodological rules, scoped hypotheses, and acceptance criteria.

Table 8: Status-coded synthesis. “Record” summarizes why the proposition is worth testing; it is not a meta-analytic effect estimate.

Proposition	Current status	Supporting record	Decisive failure or test
Causal-geometry constraint	Structural consequence under declared equations	Reachability, exact symmetry, cuts, and typed composition follow from specified graphs and quotients	A claimed hard constraint disappears after all omitted paths, shared state, readers, and granularity are modeled
Evidence Firewall	Methodological type rule	Probes, attribution, sparse codes, patching, and reconstruction license different claims and have distinct failure modes	A method-specific output predicts native use, transport, and composition without the supposedly missing evidence
Machine-Prior hypothesis	Central empirical hypothesis	Matched AR/ATR and state-tracking studies show architecture- and training-conditioned mechanism differences	A task-, resource-, behavior-, and training-only model matches MesoGenome-conditioned held-out mechanism prediction within the equivalence margin
Syntax-edit response	Central causal test	Sharing, boundary, state, position, routing, and convolution components have known structural or scoped mechanistic effects	Operationally used edits fail to move the preregistered mechanism fingerprint
Cut-pressure hypothesis	Scoped architecture hypothesis	Recurrent cuts, graph bottlenecks, and finite robust packing create distinguishability pressure under load	Increasing cut capacity or bypassing the cut does not move the predicted failure profile
Failure-interface hypothesis	Open cross-architecture hypothesis	Retrieval, compression, propagation, routing, and trajectory failures occur in architecture-linked case studies	Task and load predict failure location as well as syntax across matched systems
Routing-nonsemanticity rule	Methodological caution	Attention, gates, and expert routes identify selection structure, not a complete transformation or consumer	Router identity alone predicts held-out function as well as the full route-transform-consumer protocol
Cross-model conservation	Cross-model hypothesis	Abstract algorithms sometimes recur while timing, physical realizations, circuits, and coordinates differ	Execution-mechanism families or coordinates transfer at least as broadly as abstract algorithms under preregistered equivalence classes
Operating-regime hypothesis	Scoped extension	Recurrent state, conditional routing, and diffusion time can change the active map despite shared parameters	A single static Machine predicts cross-regime interventions and transport within tolerance
MESO SEAL	Acceptance criterion	Readability and localization can dissociate from low-collateral control; composition and transplant offer stronger tests	Readability/localization-selected accounts reconstruct and transport as reliably as accounts selected on structural holdouts

9 Decisive tests of the theory

The empirical protocol is an instrument for testing the theory, not the paper’s identity. Two results would provide the clearest initial evidence. First, a syntax-aware model should predict a mechanism fingerprint on held-out training populations better than a model given task, optimization, resource, and behavioral descriptors alone. Second, an operationally active MesoGene edit should move the mechanism fingerprint and failure-interface profile in a preregistered direction, including settings where accuracy remains equivalent. Appendix A gives the complete tasks, architecture contrasts, ground-truth regimes, data roles, estimators, power targets, and preregistration requirements.

Phase I combines three tasks—orbit transport, pointer lookup, and held-out typed composition—with four core architecture contrasts. It uses three evidential regimes: planted-mechanism organisms, trained synthetic models whose task is known but mechanism is not, and natural models with predictive and constructive rather than ontological scoring. Candidate mechanisms are nominated without holdout access and tested on unseen interventions, compositions, transports, and architecture edits.

Table 9: CTM stress–edit matrix for the first decisive experiments. Stressors challenge a role while preserving task semantics; edits alter a preregistered MesoGene. Off-diagonal responses are expected and are part of the theory.

Target role	Factorial stressors	Controlled MesoGene edits	Primary profile
Compute	Operation novelty, composition depth, withheld order, arithmetic or rule complexity	Depth, operator sharing, local transform capacity, expert/MLP allocation	$\widehat{\Sigma}_C$ and compute–memory recomputation response
Transport	Geometric/token/graph distance, topology, distractor count and similarity, route ambiguity	Local/global/dilated edges, QK/routing class, position syntax, graph rewiring	$\widehat{\Sigma}_T$ and route–transform interaction
Memory	Concurrent load, retention horizon, overwrite, noise/precision, query delay	State/cache size, eviction, bottleneck width, skip or external memory, retention dynamics	$\widehat{\Sigma}_M$ and storage–selection coupling

Let g_j denote an operationally verified edit to the nominal CTM syntax and let Σ_i be one held-out profile coordinate. The local edit-response map

$$R_{ij} = \frac{\partial \mathbb{E}[\Sigma_i]}{\partial g_j}$$

is estimated by paired training when possible and by matched population contrasts otherwise. R is not assumed diagonal: increasing memory can reduce compute by enabling cache reuse, while global transport can change which memory organization is learned. The CTM proposal earns predictive content only if preregistered blocks or signs of R generalize beyond an unstructured list of architecture features.

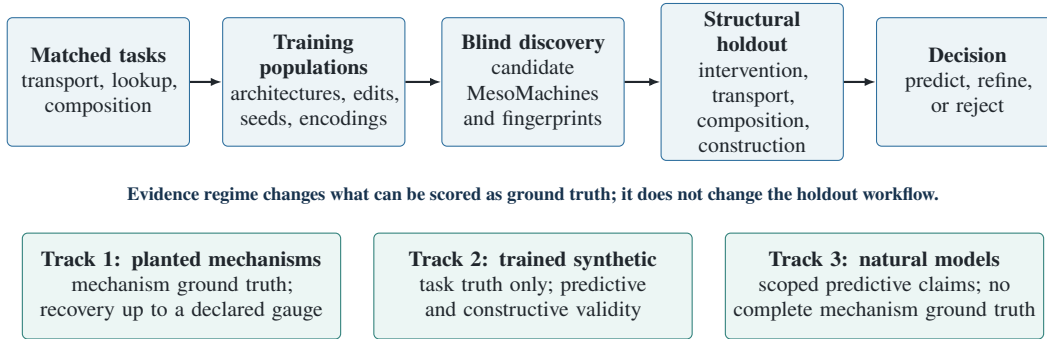


Figure 13: Decisive tests of the theory. Ground truth changes across tracks, but the logic does not: discover without holdout access, test structural predictions, and ask whether syntax and MesoGene edits add predictive value beyond task, training, behavior, and resource baselines. The top-row arrows show experimental workflow order rather than a causal graph of the model; the three lower boxes are parallel evidence regimes, not sequential stages.

The primary population-level question is the architecture-aware predictive advantage Δ_{arch} from section 3. The strongest causal question is whether a preregistered MesoGene edit moves $\sigma(M)$ in the predicted direction while operational use of the edit is verified. The strongest inverse criterion is MESO SEAL: a recovered account predicts held-out intervention and supports selective rescue, composition, transport, or reconstruction under shift. Accuracy alone is never the primary endpoint.

10 Falsification and contradiction handling

A conditional prior is not decisively refuted by one failed task unless its scope included that task. The program is weakened when preregistered syntax variables fail to improve held-out prediction repeatedly across scoped tasks, or when MesoGene edits do not move the predicted mechanism fingerprint despite being operationally used.

Table 10: Predeclared diagnosis categories. They constrain, rather than excuse, theory revision.

Category	Evidence required	Consequence
Assumption failure	Demonstrate a missing edge, broken symmetry, inadequate precision/noise premise, or unscoped reader	Narrow theorem application; do not change the empirical target
Syntax effect absent	Edit is operationally active but mechanism fingerprint stays within null/equivalence region	Lower or reject the Machine Prior
Training override	Same MesoGenome, controlled training edit changes Machine distribution	Add the training variable; reduce architecture-only claim
Method mismatch	Independent methods targeting different estimands disagree as predicted by their contracts	Keep separate licenses; do not average scores
Non-identifiability	Candidate Machines make equivalent held-out predictions under the full contract	Return both in $\widehat{\mathfrak{M}}_{\varepsilon, \delta}$
Post-hoc rescue	New Machine class or threshold introduced after holdout inspection	Exploratory only; requires a fresh holdout

The strongest failure condition is:

Across preregistered tasks and architecture edits, a task-and-training-only baseline predicts mechanism fingerprints, held-out interventions, and failure locations as well as a syntax-conditioned model within the equivalence margin.

That result would reject the program’s central predictive increment even if architecture remains useful for performance. A second failure is specific to the CTM packaging: if the typed CTM profiles and edit-response structure predict no better than the same nominal syntax variables entered as an unstructured feature list, then CTM has supplied a visualization and vocabulary but not a scientific compression. The paper treats that as a substantive negative result.

11 Discussion

11.1 The scientific claim and its novelty boundary

The Computational Mesoscale changes the dependent variable of architecture science from behavior alone to learned CTM organization. The triad itself is not a discovery: computer architecture, distributed systems, dynamical systems, and process theories have long studied computation, communication, and storage (Hong and Kung, 1981; Kissinger, 2012; Fong et al., 2015). The proposed contribution is a different object and a different burden of proof: recover the *learned, typed, causal* allocation of compute, transport, and memory inside a trained network; predict its distribution from architecture and training; and move it with controlled edits while behavior is held within an equivalence region.

Its hard content concerns paths, cuts, exact symmetries, typed composition, and robust resource bounds. Its central empirical content is the architecture- and training-conditioned Machine Prior Π^Λ . Its inverse content is a disciplined recovery problem over evidence-compatible MesoMachines. The benchmark is a falsification instrument for this theory, not the theory’s identity. The paper does not infer semantic labels from architecture, identify one Machine per model, privilege an interpretability basis, equate task ground truth with mechanism ground truth, or claim that CTM is already a complete theory of intelligence.

11.2 Typed equivalence reporting

Cross-model “sameness” is not a single relation. For a pair of systems define the *cross-model comparison map*

$$\mathbf{C}(A, B) = (c_{\text{beh}}, c_{\text{alg}}, c_{\text{obj}}, c_{\text{trk}}, c_{\text{coord}}, c_{\text{act}}),$$

where each coordinate is a separately estimated equivalence claim with its own scope, tolerance, and uncertainty. Figure 14 shows the six claim types. They are not a hierarchy or lattice: two systems may conserve an algorithm while changing physical realizations, or align coordinates while implementing different algorithms. Appendix B gives the evidence requirements for each coordinate.

Physical specificity often demands stronger evidence, but the levels are not nested in all settings. For example, two systems can have aligned coordinates without implementing the same algorithm, or implement the same algorithm

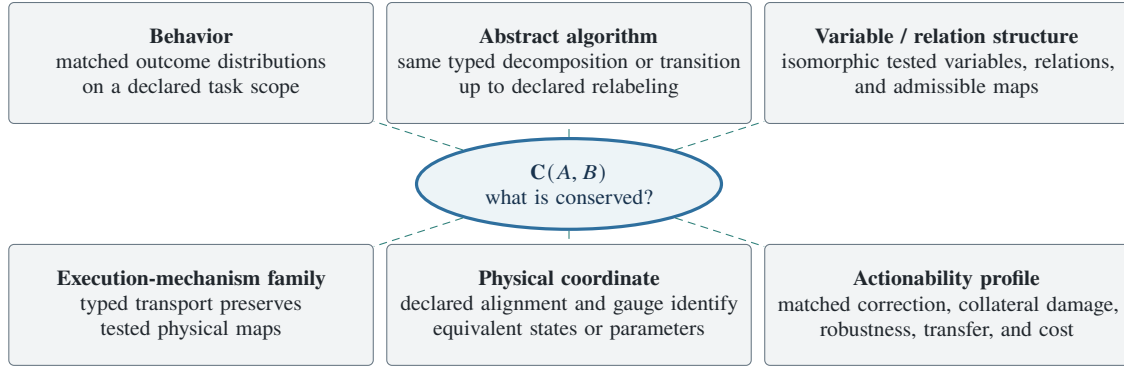


Figure 14: The cross-model comparison map. Cross-model sameness decomposes into six independently tested coordinates. The dashed spokes associate each claim with the comparison; they deliberately do not form an implication chain.

through non-transportable execution mechanism families. Every comparison must state the level, gauge, scope, and tolerance.

11.3 Why method relativity is not relativism

Method and frame indexing does not imply that all stories are equally valid. It makes disagreements testable. Candidate Machines must predict data not used to fit or select them. Increasing the intervention/readout family can refine or destroy an equivalence class. Complexity constraints prevent arbitrary nonlinear alignments from manufacturing any desired abstraction (Sutter et al., 2025). Measurement-method disagreement is evidence about target mismatch or underdetermination, not permission to choose the most appealing story.

11.4 The MESO SEAL

No mechanism earns the MESO SEAL by readability alone. A mechanism is not sealed until it can *Survive, Edit, Act, and Link*. The acronym records four independent acceptance families:

- S** **Survive unseen structure:** predict an intervention, transport, composition, or failure pattern not used for fitting or selection.
- E** **Edit predictably:** a preregistered MesoGene edit moves the mechanism fingerprint or failure interface in the predicted direction.
- A** **Act causally:** selective disable, rescue, or reconstruction changes the target with bounded collateral damage.
- L** **Link compositionally:** execution mechanisms compose on withheld cases and survive declared changes of position, context, time, or implementation.

Readability, localization, and actionability remain separate coordinates. The SEAL is therefore a profile rather than a scalar badge, summarized by the capability graph

$$\text{predict held-out intervention} \longrightarrow \left\{ \begin{array}{l} \text{selective disable/rescue,} \\ \text{withheld composition,} \\ \text{typed transport,} \\ \text{reconstruction under shift.} \end{array} \right.$$

The arrows are research goals, not logical implications. A claim never inherits a downstream evidence license without the corresponding experiment.

12 Limitations and epistemic status

The framework is not yet a calibrated predictive theory. It lacks results from the integrated empirical programme, a universal mechanism-class representation, and a validated estimator for continuous Machine families. The reachability, packing, and quotient ingredients are standard; the finite-cut versus addressable-memory law is an elementary structural separation whose scientific value depends on whether its predicted redistribution of mechanism and failure is observed in trained populations. All formal statements depend on declared structural equations, readers, tolerances, precision/noise, and graph granularity. The Machine Prior is therefore a proposed target of estimation, not a measured universal law.

CTM is not proved to be the unique, minimal, or sufficient basis for every neural mesoscale. Because roles are relational and can overlap, some physical processes may remain underidentified between compute, transport, and memory under a limited intervention family. The proposed response is to report a role profile or admissible alternatives, not to force an exclusive label. A successful CTM prediction must also beat an unstructured-syntax baseline; clean terminology alone is not empirical evidence. Selective evidence synthesis is vulnerable to publication and recency bias; many 2026 sources are preprints frozen at the version available on July 27, 2026. The evidence atlas improves traceability but does not provide dual coding, exhaustive database search, or risk-of-bias assessment.

Architecture families contain heterogeneous implementations. Mechanism classes can remain non-identifiable even after extensive intervention, and MDL partitions depend on the coding language. Natural-donor interventions can still violate support, and reconstruction may exploit handles not used natively. The proposed Phase I is intentionally narrow; success would not establish the GNN, MoE, diffusion, multimodal, or embodied extensions in Appendix C.

13 Conclusion

Neural architecture research has a missing dependent variable: the learned organization of *compute*, *transport*, and *memory* between physical neural execution and macroscopic model behavior. Architecture sets the feasible CTM trade-offs; training selects an operating organization from that landscape; interpretability faces the inverse problem of recovering it. This paper calls the nominal typed CTM syntax the MesoGenome, an editable syntax coordinate a MesoGene, the learned causal organization a MesoMachine, and its architecture- and training-conditioned distribution the Machine Prior:

$$\Pi_{S_0, \mathcal{E}}^\Lambda = \mathcal{L}\left([M_{\theta, z}^{\text{meso}}]_\Lambda\right).$$

The formal core shows what a CTM theory can already say without guessing semantics. Compute that depends on a remote state requires a transport path. Transport through a robust finite cut bounds the memory distinctions that can survive. Memory can replace recomputation only when a later native reader uses the retained memory state. Physical operations descend to abstract composition only when they preserve the declared realization equivalences. The finite-cut versus addressable-memory law makes one exchange rate explicit: a fixed robust cut pays exponentially growing distinguishability cost for independently queryable bindings, whereas extensive addressable memory grows with the records and relocates pressure to selection, delivery, and consumption. Architecture can therefore move the Machine and its failure interface before it moves the behavioral score.

The programme has a decisive acceptance criterion. Syntax must improve out-of-sample CTM prediction beyond task, training, resources, behavior, and an unstructured syntax baseline. Controlled MesoGene edits must move role profiles and their interactions in preregistered directions. Recovered Machines must predict evidence not used to construct them: interventions, compositions, transports, persistence/recomputation responses, or reconstruction under shift. If these tests fail, CTM remains useful notation but the Machine Prior adds no scientific variable beyond ordinary behavioral inductive bias. If they succeed, the forward, inverse, and edit problems become a cumulative architecture science.

This paper deliberately stops at the first closed, falsifiable layer. It does not axiomatize intelligence, solve semantics, or ground internal variables in the external world. In the spirit of Hilbert’s sixth problem, the next step is not a larger metaphor but a disciplined continuation: from neural micro-execution to a causal computational mesoscale, then from that mesoscale to structural semantics, grounding, agency, and alignment, with every bridge held to the same standards of identification, composition, transport, and constructive control. The long-term aim is to turn architecture science from a theory of possible functions into a theory of possible and probable *learned Machines*: how they form, which organizations recur across physical implementations, where they break, and how they can be rebuilt in a new model or body. The present contribution is not the completion of that programme. It is an attempt to make its first bridge precise enough to test, reject, and continue.

References

- Julius Adebayo, Justin Gilmer, Michael Muelly, et al. Sanity checks for saliency maps. In *Advances in Neural Information Processing Systems*, 2018. URL <https://arxiv.org/abs/1810.03292>.
- Uri Alon and Eran Yahav. On the bottleneck of graph neural networks and its practical implications. In *International Conference on Learning Representations*, 2021. URL <https://arxiv.org/abs/2006.05205>.
- Aryaman Arora, Neil Rathi, Nikil Roashan Selvam, Robert Csordas, Dan Jurafsky, and Christopher Potts. Mechanistic evaluation of transformers and state space models, 2025. URL <https://arxiv.org/abs/2505.15105>. arXiv:2505.15105, preprint.

- Aryaman Arora, Zhengxuan Wu, Jacob Steinhardt, and Sarah Schwettmann. Language model circuits are sparse in the neuron basis, 2026. URL <https://arxiv.org/abs/2601.22594>. arXiv:2601.22594, preprint.
- Sanjay Basu, Sadiq Y. Patel, Parth Sheth, et al. Interpretability without actionability: Mechanistic methods cannot correct language model errors despite near-perfect internal representations, 2026. URL <https://arxiv.org/abs/2603.18353>. arXiv:2603.18353, preprint.
- Nora Belrose, Zach Furman, Logan Smith, et al. Eliciting latent predictions from transformers with the tuned lens, 2023. URL <https://arxiv.org/abs/2303.08112>. arXiv:2303.08112, preprint.
- Tom Burgert, Oliver Stoll, Paolo Rota, and Begum Demir. Imagenet-trained cnns are not biased towards texture: Revisiting feature reliance through controlled suppression. In *Advances in Neural Information Processing Systems*, 2025. URL <https://proceedings.neurips.cc/>.
- David Chanin. Are sparse autoencoder benchmarks reliable?, 2026. URL <https://arxiv.org/abs/2605.18229>. arXiv:2605.18229, preprint.
- Jose A. Chavez. On the redundancy of timestep embeddings in diffusion models, 2026. URL <https://arxiv.org/abs/2606.20416>. arXiv:2606.20416, preprint.
- Bryan Cheng and Jasper Zhang. Single-position intervention fails: Distributed output templates drive in-context learning, 2026. URL <https://arxiv.org/abs/2605.04061>. arXiv:2605.04061, preprint.
- Taco Cohen and Max Welling. Group equivariant convolutional networks. In *Proceedings of the 33rd International Conference on Machine Learning*, 2016. URL <https://proceedings.mlr.press/v48/cohenc16.html>.
- Jean-Baptiste Cordonnier, Andreas Loukas, and Martin Jaggi. On the relationship between self-attention and convolutional layers, 2019. URL <https://arxiv.org/abs/1911.03584>. arXiv:1911.03584, preprint.
- Adrian Csizsarik, Peter Korosi-Szabo, Akos K. Matszangosz, Gergely Papp, and Daniel Varga. Similarity and matching of neural network representations. In *Advances in Neural Information Processing Systems*, 2021. URL <https://arxiv.org/abs/2110.14633>.
- Hoagy Cunningham, Aidan Ewart, Logan Riggs, and Robert Huben. Sparse autoencoders find highly interpretable features in language models, 2023. URL <https://arxiv.org/abs/2309.08600>. arXiv:2309.08600, preprint.
- Carlton Downey, Ahmed Hefny, Boyue Li, Byron Boots, and Geoffrey Gordon. Predictive state recurrent neural networks. In *Advances in Neural Information Processing Systems*, 2017. URL <https://arxiv.org/abs/1705.09353>.
- Yufeng Du, Minyang Tian, Srikanth Ronanki, et al. Context length alone hurts llm performance despite perfect retrieval. In *Findings of EMNLP*, 2025. URL <https://aclanthology.org/2025.findings-emnlp.1264/>.
- Nelson Elhage, Neel Nanda, Catherine Olsson, et al. A mathematical framework for transformer circuits. Transformer Circuits Thread, 2021. URL <https://transformer-circuits.pub/2021/framework/index.html>.
- Lucy Farnik, Tim Lawson, Conor Houghton, and Laurence Aitchison. Jacobian sparse autoencoders: Sparsify computations, not just activations, 2025. URL <https://arxiv.org/abs/2502.18147>. arXiv:2502.18147, preprint.
- William Fedus, Barret Zoph, and Noam Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research*, 23(120):1–39, 2022. URL <https://arxiv.org/abs/2101.03961>.
- Norm Ferns, Prakash Panangaden, and Doina Precup. Metrics for finite markov decision processes. In *Conference on Uncertainty in Artificial Intelligence*, 2004. URL <https://arxiv.org/abs/1207.4114>.
- Brendan Fong, Paolo Rapisarda, and Paweł Sobociński. A categorical approach to open and interconnected dynamical systems. *arXiv preprint arXiv:1510.05076*, 2015. URL <https://arxiv.org/abs/1510.05076>.

- Juan L. Gamella, Simon Bing, and Jakob Runge. Sanity checking causal representation learning on a simple real-world system. In *International Conference on Machine Learning*, 2025. URL <https://proceedings.mlr.press/v267/gamella25a.html>.
- Leo Gao, Tom Dupré la Tour, Henk Tillman, et al. Scaling and evaluating sparse autoencoders, 2024a. URL <https://arxiv.org/abs/2406.04093>. arXiv:2406.04093, preprint.
- Muhan Gao, TaiMing Lu, Kuai Yu, Adam Byerly, and Daniel Khashabi. Insights into llm long-context failures: When transformers know but don’t tell. In *Findings of EMNLP*, 2024b. URL <https://aclanthology.org/2024.findings-emnlp.447/>.
- Atticus Geiger, Duligur Ibeling, Amir Zur, et al. Causal abstraction: A theoretical foundation for mechanistic interpretability, 2023. URL <https://arxiv.org/abs/2301.04709>. arXiv:2301.04709, preprint.
- Robert Geirhos, Patricia Rubisch, Claudio Michaelis, et al. Imagenet-trained cnns are biased towards texture; increasing shape bias improves accuracy and robustness. In *International Conference on Learning Representations*, 2019. URL <https://openreview.net/forum?id=Bygh9j09KX>.
- Zhiren Gong, Zihao Zeng, Chau Yuen, and Wei Yang Bryan Lim. Conditional co-ablation: Recovering self-repair backups in transformer circuits, 2026. URL <https://arxiv.org/abs/2607.01940>. arXiv:2607.01940, preprint.
- Alexander N. Gorban. Hilbert’s sixth problem: The endless road to rigour. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 376(2118):20170238, 2018. doi: 10.1098/rsta.2017.0238. URL <https://doi.org/10.1098/rsta.2017.0238>.
- Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces, 2023. URL <https://arxiv.org/abs/2312.00752>. arXiv:2312.00752, preprint.
- Albert Gu, Karan Goel, and Christopher Ré. Efficiently modeling long sequences with structured state spaces, 2021. URL <https://arxiv.org/abs/2111.00396>. arXiv:2111.00396, preprint.
- Adi Haviv, Ori Ram, Ofir Press, Peter Izsak, and Omer Levy. Transformer language models without positional encodings still learn positional information. In *Findings of EMNLP*, 2022. URL <https://aclanthology.org/2022.findings-emnlp.99/>.
- Thomas Heap, Tim Lawson, Lucy Farnik, and Laurence Aitchison. Sparse autoencoders can interpret randomly initialized transformers, 2025. URL <https://arxiv.org/abs/2501.17727>. arXiv:2501.17727, preprint.
- Katherine Hermann, Ting Chen, and Simon Kornblith. The origins and prevalence of texture bias in convolutional neural networks. In *Advances in Neural Information Processing Systems*, 2020. URL <https://proceedings.neurips.cc/>.
- John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In *EMNLP-IJCNLP*, 2019. URL <https://aclanthology.org/D19-1275/>.
- David Hilbert. Mathematical problems. *Bulletin of the American Mathematical Society*, 8:437–479, 1902. doi: 10.1090/S0002-9904-1902-00923-3. URL <https://doi.org/10.1090/S0002-9904-1902-00923-3>.
- Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997.
- Jia-Wei Hong and H. T. Kung. I/o complexity: The red-blue pebble game. In *Proceedings of the Thirteenth Annual ACM Symposium on Theory of Computing*, pages 326–333, 1981. URL <https://www.eecs.harvard.edu/~htk/publication/1981-stoc-hong-kung.pdf>.
- Md Amirul Islam, Matthew Kowal, Sen Jia, Konstantinos G. Derpanis, and Neil D. B. Bruce. Position, padding and predictions: A deeper look at position information in cnns. In *CVPR*, 2021. URL <https://arxiv.org/abs/2101.12322>.
- Sarthak Jain and Byron C. Wallace. Attention is not explanation. In *NAACL*, 2019. URL <https://aclanthology.org/N19-1357/>.

- Adarsh Jamadandi, Celia Rubio-Madrigal, and Rebekka Burkholz. Spectral graph pruning against over-squashing and over-smoothing. In *Advances in Neural Information Processing Systems*, 2024. URL <https://proceedings.neurips.cc/>.
- Samy Jelassi, David Brandfonbrener, Sham M. Kakade, and Eran Malach. Repeat after me: Transformers are better than state space models at copying. In *International Conference on Machine Learning*, 2024. URL <https://proceedings.mlr.press/v235/jelassi24a.html>.
- Ahmed Karim, Fatima SheaiB, Zein Khamis, et al. Attention deficits in language models: Causal explanations for procedural hallucinations, 2026. URL <https://arxiv.org/abs/2602.19239>. arXiv:2602.19239, preprint.
- Amirhossein Kazemnejad, Inkit Padhi, Karthikeyan Natesan Ramamurthy, Payel Das, and Siva Reddy. The impact of positional encoding on length generalization in transformers, 2023. URL <https://arxiv.org/abs/2305.19466>. arXiv:2305.19466, preprint.
- Ilyes Khemakhem, Diederik P. Kingma, Ricardo Monti, and Aapo Hyvärinen. Variational autoencoders and nonlinear ica: A unifying framework. In *AISTATS*, 2020. URL <https://arxiv.org/abs/1907.04809>.
- Aleks Kissinger. *Pictures of Processes: Automated Graph Rewriting for Monoidal Categories and Applications to Quantum Computing*. PhD thesis, University of Oxford, 2012. URL <https://arxiv.org/abs/1203.0202>.
- Belinda Z. Li, Zifan Carl Guo, and Jacob Andreas. (how) do language models track state? In *International Conference on Machine Learning*, 2025. URL <https://proceedings.mlr.press/v267/li25r.html>.
- Shanda Li, Xiangning Chen, Di He, and Cho-Jui Hsieh. Can vision transformers perform convolution?, 2021. URL <https://arxiv.org/abs/2111.01353>. arXiv:2111.01353, preprint.
- Nelson F. Liu, Kevin Lin, John Hewitt, et al. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173, 2024. URL <https://aclanthology.org/2024.tacl-1.9/>.
- Francesco Locatello, Stefan Bauer, Mario Lucic, et al. Challenging common assumptions in the unsupervised learning of disentangled representations. In *International Conference on Machine Learning*, 2019. URL <https://arxiv.org/abs/1811.12359>.
- Grace Luo, Lisa Dunlap, Dong Huk Park, Aleksander Holynski, and Trevor Darrell. Diffusion hyperfeatures: Searching through time and space for semantic correspondence. In *Advances in Neural Information Processing Systems*, 2023. URL <https://proceedings.neurips.cc/>.
- Wenjie Luo, Yujia Li, Raquel Urtasun, and Richard Zemel. Understanding the effective receptive field in deep convolutional neural networks. In *Advances in Neural Information Processing Systems*, 2016. URL <https://proceedings.neurips.cc/>.
- Maxime Méloux, Silviu Maniu, François Portet, and Maxime Peyrard. Everything, everywhere, all at once: Is mechanistic interpretability identifiable?, 2025. URL <https://arxiv.org/abs/2502.20914>. arXiv:2502.20914, preprint.
- William Merrill, Jackson Petty, and Ashish Sabharwal. The illusion of state in state-space models. In *International Conference on Machine Learning*, 2024. URL <https://proceedings.mlr.press/v235/merrill24a.html>.
- Jack Merullo, Carsten Eickhoff, and Ellie Pavlick. Circuit component reuse across tasks in transformer language models. In *International Conference on Learning Representations*, 2024. URL <https://proceedings.iclr.cc/>.
- Aaron Mueller, Atticus Geiger, Sarah Wiegreffe, et al. Mib: A mechanistic interpretability benchmark, 2025. URL <https://arxiv.org/abs/2504.13151>. arXiv:2504.13151, preprint.
- Anil Nerode. Linear automaton transformations. *Proceedings of the American Mathematical Society*, 9(4):541–544, 1958.

- Catherine Olsson, Nelson Elhage, Neel Nanda, et al. In-context learning and induction heads. Transformer Circuits Thread, 2022. URL <https://transformer-circuits.pub/2022/in-context-learning-and-induction-heads/>.
- Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2 edition, 2009.
- William Peebles and Saining Xie. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023. URL <https://arxiv.org/abs/2212.09748>.
- Theodora-Mara Pîslar, Sara Magliacane, and Atticus Geiger. Combining causal models for more accurate abstractions of neural networks. In *Conference on Causal Learning and Reasoning*, 2025. URL <https://proceedings.mlr.press/v275/pislar25a.html>.
- Siamak Ravanbakhsh. Universal equivariant multilayer perceptrons. In *International Conference on Machine Learning*, 2020. URL <https://proceedings.mlr.press/v119/ravanbakhsh20a.html>.
- Abhilasha Ravichander, Yonatan Belinkov, and Eduard Hovy. Probing the probing paradigm: Does probing accuracy entail task relevance? In *EACL*, 2021. URL <https://aclanthology.org/2021.eacl-main.143/>.
- Anian Ruoss, Grégoire Delétang, Tim Genewein, et al. Randomized positional encodings boost length generalization of transformers. In *ACL*, 2023. URL <https://aclanthology.org/2023.acl-short.161/>.
- Valeria Ruscio, Umberto Nanni, and Fabrizio Silvestri. Where does absolute position come from in decoder-only transformers?, 2026. URL <https://arxiv.org/abs/2606.06160>. arXiv:2606.06160, preprint.
- Bernhard Schölkopf, Francesco Locatello, Stefan Bauer, et al. Toward causal representation learning. *Proceedings of the IEEE*, 2021. URL <https://arxiv.org/abs/2102.11107>.
- Tamar Rott Shaham, Sarah Schwettmann, Franklin Wang, et al. A multimodal automated interpretability agent. In *International Conference on Machine Learning*, 2024. URL <https://proceedings.mlr.press/v235/shaham24a.html>.
- Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarski, et al. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In *International Conference on Learning Representations*, 2017. URL <https://arxiv.org/abs/1701.06538>.
- Satinder Singh, Michael James, and Matthew Rudary. Predictive state representations: A new theory for modeling dynamical systems, 2012. URL <https://arxiv.org/abs/1207.4167>. arXiv:1207.4167.
- Koustuv Sinha, Amirhossein Kazemnejad, Siva Reddy, et al. The curious case of absolute position embeddings. In *Findings of EMNLP*, 2022. URL <https://aclanthology.org/2022.findings-emnlp.326/>.
- Denis Sutter, Julian Minder, Thomas Hofmann, and Tiago Pimentel. The non-linear representation dilemma: Is causal abstraction enough for mechanistic interpretability?, 2025. URL <https://arxiv.org/abs/2507.08802>. arXiv:2507.08802, preprint.
- Luming Tang, Menglin Jia, Qianqian Wang, Cheng Perng Phoo, and Bharath Hariharan. Emergent correspondence from image diffusion. In *Advances in Neural Information Processing Systems*, 2023. URL <https://proceedings.neurips.cc/>.
- Asher Trockman, Hrayr Harutyunyan, J. Zico Kolter, Sanjiv Kumar, and Srinadh Bhojanapalli. Mimetic initialization helps state space models learn to recall, 2024. URL <https://arxiv.org/abs/2410.11135>. arXiv:2410.11135, preprint.
- Sankaran Vaidyanathan, David Arbour, Aaron Mueller, Scott Niekum, and David Jensen. The curse of multiple mediators: Hidden interaction effects in activation patching, 2026. URL <https://arxiv.org/abs/2606.27510>. arXiv:2606.27510, preprint.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, et al. Attention is all you need. In *Advances in Neural Information Processing Systems*, 2017. URL <https://arxiv.org/abs/1706.03762>.

- Arun Venkatraman, Nicholas Rhinehart, Wen Sun, et al. Predictive-state decoders: Encoding the future into recurrent networks, 2017. URL <https://arxiv.org/abs/1709.08520>. arXiv:1709.08520, preprint.
- Binxu Wang, Jingxuan Fan, and Xu Pan. Circuit mechanisms for spatial relation generation in diffusion transformers, 2026. URL <https://arxiv.org/abs/2601.06338>. arXiv:2601.06338, preprint.
- Junxuan Wang, Xuyang Ge, Wentao Shu, et al. Towards universality: Studying mechanistic similarity across language model architectures, 2024. URL <https://arxiv.org/abs/2410.06672>. arXiv:2410.06672, preprint.
- Shida Wang and Qianxiao Li. Stablessm: Alleviating the curse of memory in state-space models through stable reparameterization. In *International Conference on Machine Learning*, 2024. URL <https://proceedings.mlr.press/v235/wang24ag.html>.
- Ziyu Wang, Wenhao Jiang, Yiming M. Zhu, et al. Dynamixer: A vision mlp architecture with dynamic mixing. In *International Conference on Machine Learning*, 2022. URL <https://proceedings.mlr.press/v162/wang22i.html>.
- Sarah Wiegrefe and Yuval Pinter. Attention is not not explanation. In *EMNLP-IJCNLP*, 2019. URL <https://aclanthology.org/D19-1002/>.
- Xinyi Wu, Amir Ajorlou, Zihui Wu, and Ali Jadbabaie. Demystifying oversmoothing in attention-based graph neural networks. In *Advances in Neural Information Processing Systems*, 2023. URL <https://proceedings.neurips.cc/>.
- Yiwei Wu, Atticus Geiger, and Raphaël Millière. How do transformers learn variable binding in symbolic programs? In *International Conference on Machine Learning*, 2025a. URL <https://proceedings.mlr.press/v267/wu25j.html>.
- Zhengxuan Wu, Aryaman Arora, Atticus Geiger, et al. Axbench: Steering llms? even simple baselines outperform sparse autoencoders, 2025b. URL <https://arxiv.org/abs/2501.17148>. arXiv:2501.17148, preprint.
- Keyulu Xu, Weihua Hu, Jure Leskovec, and Stefanie Jegelka. How powerful are graph neural networks?, 2018. URL <https://arxiv.org/abs/1810.00826>. arXiv:1810.00826.
- Fuzhao Xue, Zian Zheng, Yao Fu, et al. Openmoe: An early effort on open mixture-of-experts language models. In *International Conference on Machine Learning*, 2024. URL <https://proceedings.mlr.press/v235/xue24c.html>.
- Kayo Yin and Jacob Steinhardt. Which attention heads matter for in-context learning? In *International Conference on Machine Learning*, 2025. URL <https://proceedings.mlr.press/v267/yin25e.html>.
- Tan Yu, Xu Li, Yunfeng Cai, Mingming Sun, and Ping Li. S2-mlp: Spatial-shift mlp architecture for vision, 2021. URL <https://arxiv.org/abs/2106.07477>. arXiv:2106.07477, preprint.
- Ziqian Zhong, Ziming Liu, Max Tegmark, and Jacob Andreas. The clock and the pizza: Two stories in mechanistic explanation of neural networks. In *Advances in Neural Information Processing Systems*, 2023. URL <https://proceedings.neurips.cc/>.

A Complete Phase I empirical protocol

This appendix makes the empirical programme operational while keeping it subordinate to the theoretical claim. Phase I uses three tasks, four core architecture contrasts, three evidence regimes, and three method families. GNN, MoE, diffusion, multimodal, and embodied extensions enter only after the core estimators and failure criteria are calibrated.

A.1 Tasks and controlled factors

1. **Orbit transport.** The same abstract object appears under a known position or group action. The primary factors are seen/unseen group element, coordinate encoding, boundary condition, and parameter tying.
2. **Pointer lookup.** A query selects one of several matched records. Absolute position, relative distance, context length, candidate count, candidate similarity, and position syntax are varied factorially rather than collapsed into “retrieval difficulty.”
3. **Held-out composition.** Primitive typed maps $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are trained, while selected instances of $g \circ f$ are withheld. Primitive success is separated from physical compositional closure.

Task encodings are randomized across coordinate systems and counterbalanced so that a grid, sequence, or graph presentation does not trivially reveal the intended mechanism. Encoding is an explicit factor, not hidden preprocessing.

A.2 Architecture contrasts and resources

Core contrasts are:

1. untied versus parameter-tied MLP controls;
2. symmetry-preserving versus symmetry-breaking CNNs, with boundary and dilation edits;
3. RNN, linear SSM, and selective SSM variants with state-capacity, selectivity, and short-convolution edits;
4. Transformer variants with preregistered attention mask and position syntax.

Two analyses are reported. A resource-matched design declares one primary budget and reports all unmatched resources; an architecture-optimized analysis compares Pareto frontiers. Parameters, FLOPs, memory state, active compute, training time, and wall time are never claimed to be simultaneously matched.

A.3 Three evidence regimes

1. **Planted-mechanism organisms.** Physical variables, execution mechanisms, redundancy, repair pathways, and intervention ground truth are constructed. Recovery is scored against mechanism ground truth up to a declared gauge.
2. **Trained synthetic models.** Only the task generator is known. Internal mechanism is unknown. Scoring uses held-out interventional prediction, MDL/compression under a declared language, composition, transport, and selective reconstruction. Task objects are not scored as “true internal objects.”
3. **Natural models.** Claims are limited to predictive validity, typed equivalence, and actionability on declared distributions. No complete mechanism ground truth or total ontology recovery is asserted.

A.4 Methods, data roles, and search budgets

Phase I uses three method families: a controlled decoder/probe, natural-donor causal intervention with interaction-aware coalitions, and reconstruction/transplant. Each receives a transparent compute, supervision, intervention, and search budget appropriate to its operation; equal access is not assumed.

The four data roles are disjoint:

$\mathcal{D}_{\text{fit}} \rightarrow$ fit sensors and alignments, $\mathcal{D}_{\text{sel}} \rightarrow$ select candidates and hyperparameters,

$\mathcal{D}_{\text{eval}} \rightarrow$ estimate preregistered effects, $\mathcal{D}_{\text{hold}} \rightarrow$ test structural generalization.

Automated search trajectories, donor optimization, prompt selection, stopping decisions, and candidate partitions belong to fitting or selection. Using them as confirmatory evidence requires selective-inference correction or a fresh holdout.

A.5 Primary tests

The central population target is whether MesoGenome information improves out-of-sample prediction of a Machine class or mechanism fingerprint:

$$\begin{aligned} \mathcal{M}_0 : \quad y &\sim p(y \mid \mathcal{T}, \mathcal{E}, \text{Res}, \text{Behavior}), \\ \mathcal{M}_1 : \quad y &\sim p(y \mid \mathcal{T}, \mathcal{E}, \text{Res}, \text{Behavior}, S_0). \end{aligned}$$

The primary comparison is held-out proper-score improvement on held-out architecture edits and task encodings. In planted organisms y may be a Machine class. In trained models it is a preregistered mechanism-fingerprint vector or a complexity-controlled partition selected without holdout access.

Power is simulated for the minimum architecture-aware score improvement, equivalence margins, edit effects, and key interactions. Confirmatory analyses freeze alpha, power, censoring, multiple-testing correction, and immutable stopping. A candidate class added after holdout inspection is exploratory and requires a new holdout.

Table 11: Phase I confirmatory skeleton. Numerical margins are pilot-dependent and must be frozen before the confirmatory run.

Question	Primary estimand	Required preregistration
Does syntax predict mechanism?	Held-out proper-score difference $\mathcal{M}_1 - \mathcal{M}_0$	Minimum relevant improvement, folds, hierarchy, seeds, stopping
Does tying improve transport?	Paired transport-defect difference	Equivalence/superiority margin, group law, donor support
Does a cut fail under load?	Capacity profile $\mathbf{I}(K)$ and packing/MI bound	Noise, precision, margin, K grid, reader class
Does address compete?	Position-matched effects of count and similarity	Factorial contrasts, position family, length, multiplicity
Do primitives compose?	Withheld Δ_o	Composite family, type map, nulls, effect margin
Does a MesoGene edit move the Machine distribution?	Paired or population edit effect on $\sigma(\mathbf{M})$	Coupling assumptions, operational-use check, direction, margin

Table 12: Equivalence-reporting types and minimum discriminating evidence. The rows are not a total order.

Declared type	Equality/equivalence asserted	Minimum discriminating evidence
Behavior	Matched outcome distribution on a declared task scope	Held-out behavioral equivalence test with uncertainty
Abstract algorithm	Same typed input–output decomposition or state transition up to declared relabeling	Competing algorithm models and held-out intermediate counterfactuals
Object/relation structure	Isomorphic recovered physical realization of objects, relations, and admissible maps	Natural-donor interchange, alternative partitions, and quotient defects
Execution-mechanism family	Typed partial transport between physical mechanism families preserves tested maps	Cross-position/model transport plus mechanism-level interventions
Physical coordinate	A declared alignment identifies equivalent states or parameters	Gauge specification, identifiability contract, and held-out coordinate transport
Actionability profile	Comparable correction, collateral damage, robustness, transfer, and cost	Matched constructive intervention under distribution shift

A.6 Acceptance and rejection

The central claim is supported only if:

1. syntax improves held-out mechanism prediction beyond task, training, resource, behavior, and encoding baselines;
2. operationally used MesoGene edits move mechanism fingerprints in preregistered directions;
3. recovered candidates predict structural holdouts not used for fitting or selection;
4. effects replicate across seeds and at least one new task encoding.

It is rejected on the tested scope if architecture-aware prediction remains within the equivalence margin of the baseline and MesoGene edits fail to move mechanisms despite verified operational use.

B Detailed equivalence evidence requirements

Physical specificity often demands stronger evidence, but the types are not nested in all settings. Two systems can have aligned coordinates without implementing the same algorithm, or implement the same algorithm through non-transportable execution mechanism families. Every comparison states type, gauge, scope, tolerance, and uncovered mass.

C Scoped architecture extensions

C.1 Message-passing GNNs

For a specific message-passing class with declared initial features, L layers expose at most the L -hop neighborhood, and expressivity can be bounded by the corresponding Weisfeiler-Lehman test (Xu et al., 2018). Node IDs, positional encodings, higher-order states, rewiring, or a global token change the class. Finite width does not imply finite distinguishability without the robust assumptions of [proposition 18](#).

Over-squashing, oversmoothing, WL ambiguity, and optimization failure are separate targets. Over-squashing is a

cut/information-pressure hypothesis; oversmoothing is contraction or homogenization; WL ambiguity is an expressivity equivalence; optimization is neither (Alon and Yahav, 2021; Jamadandi et al., 2024; Wu et al., 2023). Input graph topology is not automatically the model’s causal information topology. Rewiring/global edges are decisive edits only if their operational use is verified.

C.2 Sparse mixtures of experts

For N experts, top- k routing, and L layers, expert identities, active sets, weighted mixtures, and cross-layer handoffs are distinct objects. The number of active-set sequences is at most $\binom{N}{k}^L$; individual branch combinations can scale as k^L within a path. Neither expression is a semantic module count. Dropping is implementation-dependent: dropless, expert-choice, rerouting, and capacity-limited systems have different cuts (Shazeer et al., 2017; Fedus et al., 2022; Xue et al., 2024).

The candidate execution mechanism is

hidden geometry $\rightarrow$ router decision $\rightarrow$ weighted expert set/path $\rightarrow$ consumer.

Expert identity alone is not a skill label. Discrete routing adaptivity is not placed on the same scalar as continuous attention adaptivity. The extension should compare expert, active-set, router-logit, full-path, and consumer-conditioned predictors with separate estimands.

C.3 Diffusion models and DiTs

Four notions must be separated.

1. A network $v_\theta(x, t, c)$ is non-autonomous when the update map depends explicitly on time t .
2. A sampler can be non-autonomous through a time-dependent noise/solver schedule even if the network omits t .
3. An autonomous map applied to changing state distributions is distributionally nonstationary, not mathematically non-autonomous.
4. Shared weights can yield different effective behavior across state regimes without changing the underlying autonomous map.

Time-embedding redundancy in some regimes is therefore a boundary on, not evidence for, universal explicit-clock use (Chavez, 2026).

Layer-by-time features and externally useful diffusion descriptors show recoverable dynamics, not native variable identity (Luo et al., 2023; Tang et al., 2023). A transport defect across timesteps must separate alignment error, sampling randomness, and solver error before being called path dependence. A stable family of operating regimes uses the cross-input coherence criteria of section 2.5.

A naive “CNN versus ViT” by “supervised versus diffusion” table is not a matched factorial, because U-Net and DiT differ in skips, conditioning, normalization, scale, and cross-attention. The appropriate design edits one component at a time or builds matched denoisers with the same conditioning and solver. DiT-specific evidence about interface-dependent circuits is currently scoped to controlled settings (Peebles and Xie, 2023; Wang et al., 2026).

C.4 External validity

Multimodal, embodied, speech, time-series, and automated-interpretability systems are future validity domains, not part of the core evidence. Closed-loop agents additionally require environment dynamics, feedback policies, safety constraints, and intervention-induced distribution shift in $\mathfrak{F}$ and $\mathfrak{C}_{\text{do}}$. Automated search additionally requires selective-inference correction because the tool adaptively chooses hypotheses, prompts, donors, and stopping times (Shaham et al., 2024).

D Preregistration minimum

Before a confirmatory run, freeze:

1. task generators, encoding randomization, data splits, software, model versions, and seeds;
2. syntax variants and the primary resource budget;
3. method classes, search budgets, donor support, and stopping rules;
4. mechanism equivalence levels and candidate partitions;
5. all estimands, effect/equivalence margins, nulls, power simulations, multiplicity correction, and censoring rules;
6. architecture-edit predictions and contradiction categories;
7. held-out interventions, composites, contexts, and transport transformations;
8. a signed timestamp or immutable artifact hash.

E Evidence-file schema

The source package contains:

- `evidence_atlas.csv`: one row per cited source-claim pair with evidence axes, status, version date, scope, and limitation;
- `claim_to_source.csv`: manuscript claim identifiers and supporting/limiting sources;
- `search_protocol.md`: the non-systematic selective-synthesis protocol and freeze date;
- `exclusion_log.csv`: sources considered during revision but excluded from evidential support or moved to extensions.

These files support auditability without claiming exhaustive retrieval or dual-coded systematic review.